

mlp_report_executed

October 5, 2026

1 CS324 Deep Learning - Assignment 1

This notebook reports the NumPy MLP experiments for Parts II and III. The implementation is imported from `train_mlp_numpy.py`; the neural-network code is not duplicated here.

```
[1]: import sys
      from pathlib import Path

      import matplotlib.pyplot as plt

      sys.path.append(str(Path.cwd()))
      from train_mlp_numpy import train
```

1.1 Part II: batch gradient descent

The default setting uses one full training-set batch per epoch.

```
[2]: batch_model, batch_history = train(
      dnn_hidden_units="20",
      learning_rate=0.01,
      max_steps=1500,
      eval_freq=10,
      batch_size=None,
      random_seed=42,
      )
```

Step: 0, Train Loss: 0.6929, Train Accuracy: 79.50%, Loss: 0.6929, Accuracy: 82.00%

Step: 10, Train Loss: 0.6929, Train Accuracy: 80.12%, Loss: 0.6929, Accuracy: 83.00%

Step: 20, Train Loss: 0.6929, Train Accuracy: 80.38%, Loss: 0.6929, Accuracy: 83.50%

Step: 30, Train Loss: 0.6929, Train Accuracy: 80.62%, Loss: 0.6928, Accuracy: 83.00%

Step: 40, Train Loss: 0.6928, Train Accuracy: 80.50%, Loss: 0.6928, Accuracy: 83.50%

Step: 50, Train Loss: 0.6928, Train Accuracy: 80.75%, Loss: 0.6928, Accuracy: 84.00%

Step: 60, Train Loss: 0.6928, Train Accuracy: 81.00%, Loss: 0.6927, Accuracy:

84.00%

Step: 70, Train Loss: 0.6927, Train Accuracy: 81.00%, Loss: 0.6927, Accuracy: 84.00%

Step: 80, Train Loss: 0.6927, Train Accuracy: 80.88%, Loss: 0.6926, Accuracy: 84.00%

Step: 90, Train Loss: 0.6926, Train Accuracy: 80.88%, Loss: 0.6926, Accuracy: 84.00%

Step: 100, Train Loss: 0.6926, Train Accuracy: 81.00%, Loss: 0.6926, Accuracy: 84.00%

Step: 110, Train Loss: 0.6926, Train Accuracy: 80.88%, Loss: 0.6925, Accuracy: 84.00%

Step: 120, Train Loss: 0.6925, Train Accuracy: 80.75%, Loss: 0.6925, Accuracy: 83.50%

Step: 130, Train Loss: 0.6925, Train Accuracy: 80.50%, Loss: 0.6924, Accuracy: 81.50%

Step: 140, Train Loss: 0.6924, Train Accuracy: 80.62%, Loss: 0.6924, Accuracy: 82.00%

Step: 150, Train Loss: 0.6924, Train Accuracy: 80.25%, Loss: 0.6923, Accuracy: 81.50%

Step: 160, Train Loss: 0.6923, Train Accuracy: 79.75%, Loss: 0.6922, Accuracy: 81.00%

Step: 170, Train Loss: 0.6922, Train Accuracy: 79.38%, Loss: 0.6922, Accuracy: 81.00%

Step: 180, Train Loss: 0.6922, Train Accuracy: 79.25%, Loss: 0.6921, Accuracy: 80.00%

Step: 190, Train Loss: 0.6921, Train Accuracy: 79.12%, Loss: 0.6920, Accuracy: 79.50%

Step: 200, Train Loss: 0.6920, Train Accuracy: 79.00%, Loss: 0.6919, Accuracy: 79.50%

Step: 210, Train Loss: 0.6920, Train Accuracy: 79.00%, Loss: 0.6919, Accuracy: 79.50%

Step: 220, Train Loss: 0.6919, Train Accuracy: 79.00%, Loss: 0.6918, Accuracy: 79.50%

Step: 230, Train Loss: 0.6918, Train Accuracy: 79.00%, Loss: 0.6917, Accuracy: 79.50%

Step: 240, Train Loss: 0.6917, Train Accuracy: 79.00%, Loss: 0.6916, Accuracy: 79.50%

Step: 250, Train Loss: 0.6916, Train Accuracy: 79.00%, Loss: 0.6915, Accuracy: 79.00%

Step: 260, Train Loss: 0.6915, Train Accuracy: 79.00%, Loss: 0.6914, Accuracy: 79.00%

Step: 270, Train Loss: 0.6914, Train Accuracy: 79.00%, Loss: 0.6912, Accuracy: 79.00%

Step: 280, Train Loss: 0.6912, Train Accuracy: 79.00%, Loss: 0.6911, Accuracy: 79.00%

Step: 290, Train Loss: 0.6911, Train Accuracy: 79.00%, Loss: 0.6910, Accuracy: 79.00%

Step: 300, Train Loss: 0.6910, Train Accuracy: 79.00%, Loss: 0.6908, Accuracy:

79.00%

Step: 310, Train Loss: 0.6908, Train Accuracy: 79.00%, Loss: 0.6907, Accuracy: 79.00%

Step: 320, Train Loss: 0.6907, Train Accuracy: 79.12%, Loss: 0.6905, Accuracy: 79.00%

Step: 330, Train Loss: 0.6905, Train Accuracy: 79.12%, Loss: 0.6903, Accuracy: 78.50%

Step: 340, Train Loss: 0.6903, Train Accuracy: 79.12%, Loss: 0.6901, Accuracy: 78.00%

Step: 350, Train Loss: 0.6901, Train Accuracy: 79.12%, Loss: 0.6899, Accuracy: 78.00%

Step: 360, Train Loss: 0.6899, Train Accuracy: 79.12%, Loss: 0.6897, Accuracy: 78.00%

Step: 370, Train Loss: 0.6897, Train Accuracy: 79.12%, Loss: 0.6895, Accuracy: 78.50%

Step: 380, Train Loss: 0.6895, Train Accuracy: 79.12%, Loss: 0.6892, Accuracy: 78.50%

Step: 390, Train Loss: 0.6893, Train Accuracy: 79.12%, Loss: 0.6890, Accuracy: 78.50%

Step: 400, Train Loss: 0.6890, Train Accuracy: 79.12%, Loss: 0.6887, Accuracy: 78.50%

Step: 410, Train Loss: 0.6887, Train Accuracy: 79.12%, Loss: 0.6884, Accuracy: 78.50%

Step: 420, Train Loss: 0.6884, Train Accuracy: 79.12%, Loss: 0.6881, Accuracy: 78.00%

Step: 430, Train Loss: 0.6881, Train Accuracy: 79.12%, Loss: 0.6877, Accuracy: 78.00%

Step: 440, Train Loss: 0.6878, Train Accuracy: 79.12%, Loss: 0.6874, Accuracy: 78.00%

Step: 450, Train Loss: 0.6874, Train Accuracy: 79.12%, Loss: 0.6870, Accuracy: 78.00%

Step: 460, Train Loss: 0.6870, Train Accuracy: 79.12%, Loss: 0.6866, Accuracy: 78.00%

Step: 470, Train Loss: 0.6866, Train Accuracy: 79.12%, Loss: 0.6862, Accuracy: 78.00%

Step: 480, Train Loss: 0.6862, Train Accuracy: 79.12%, Loss: 0.6857, Accuracy: 78.00%

Step: 490, Train Loss: 0.6858, Train Accuracy: 79.12%, Loss: 0.6852, Accuracy: 78.00%

Step: 500, Train Loss: 0.6853, Train Accuracy: 79.12%, Loss: 0.6847, Accuracy: 78.00%

Step: 510, Train Loss: 0.6848, Train Accuracy: 79.12%, Loss: 0.6842, Accuracy: 78.00%

Step: 520, Train Loss: 0.6842, Train Accuracy: 79.12%, Loss: 0.6836, Accuracy: 78.00%

Step: 530, Train Loss: 0.6836, Train Accuracy: 79.12%, Loss: 0.6829, Accuracy: 78.00%

Step: 540, Train Loss: 0.6830, Train Accuracy: 79.12%, Loss: 0.6823, Accuracy:

78.00%

Step: 550, Train Loss: 0.6824, Train Accuracy: 79.12%, Loss: 0.6816, Accuracy: 78.00%

Step: 560, Train Loss: 0.6817, Train Accuracy: 79.12%, Loss: 0.6808, Accuracy: 78.00%

Step: 570, Train Loss: 0.6810, Train Accuracy: 79.12%, Loss: 0.6801, Accuracy: 78.00%

Step: 580, Train Loss: 0.6802, Train Accuracy: 79.12%, Loss: 0.6792, Accuracy: 78.00%

Step: 590, Train Loss: 0.6794, Train Accuracy: 79.12%, Loss: 0.6783, Accuracy: 78.00%

Step: 600, Train Loss: 0.6785, Train Accuracy: 79.12%, Loss: 0.6774, Accuracy: 78.00%

Step: 610, Train Loss: 0.6776, Train Accuracy: 79.12%, Loss: 0.6764, Accuracy: 78.00%

Step: 620, Train Loss: 0.6766, Train Accuracy: 79.12%, Loss: 0.6754, Accuracy: 78.00%

Step: 630, Train Loss: 0.6756, Train Accuracy: 79.12%, Loss: 0.6742, Accuracy: 78.00%

Step: 640, Train Loss: 0.6745, Train Accuracy: 79.12%, Loss: 0.6731, Accuracy: 78.00%

Step: 650, Train Loss: 0.6733, Train Accuracy: 79.12%, Loss: 0.6718, Accuracy: 78.00%

Step: 660, Train Loss: 0.6721, Train Accuracy: 79.12%, Loss: 0.6705, Accuracy: 78.00%

Step: 670, Train Loss: 0.6708, Train Accuracy: 79.12%, Loss: 0.6691, Accuracy: 78.00%

Step: 680, Train Loss: 0.6695, Train Accuracy: 79.12%, Loss: 0.6677, Accuracy: 78.00%

Step: 690, Train Loss: 0.6680, Train Accuracy: 79.12%, Loss: 0.6662, Accuracy: 78.00%

Step: 700, Train Loss: 0.6665, Train Accuracy: 79.12%, Loss: 0.6646, Accuracy: 78.00%

Step: 710, Train Loss: 0.6650, Train Accuracy: 79.12%, Loss: 0.6629, Accuracy: 78.00%

Step: 720, Train Loss: 0.6633, Train Accuracy: 79.12%, Loss: 0.6611, Accuracy: 78.00%

Step: 730, Train Loss: 0.6616, Train Accuracy: 79.12%, Loss: 0.6592, Accuracy: 78.00%

Step: 740, Train Loss: 0.6598, Train Accuracy: 79.12%, Loss: 0.6572, Accuracy: 78.50%

Step: 750, Train Loss: 0.6579, Train Accuracy: 79.12%, Loss: 0.6552, Accuracy: 78.50%

Step: 760, Train Loss: 0.6559, Train Accuracy: 79.12%, Loss: 0.6530, Accuracy: 78.50%

Step: 770, Train Loss: 0.6538, Train Accuracy: 79.12%, Loss: 0.6508, Accuracy: 78.50%

Step: 780, Train Loss: 0.6516, Train Accuracy: 79.12%, Loss: 0.6484, Accuracy:

78.50%

Step: 790, Train Loss: 0.6493, Train Accuracy: 79.12%, Loss: 0.6459, Accuracy: 78.50%

Step: 800, Train Loss: 0.6470, Train Accuracy: 79.12%, Loss: 0.6434, Accuracy: 78.50%

Step: 810, Train Loss: 0.6445, Train Accuracy: 79.12%, Loss: 0.6407, Accuracy: 78.00%

Step: 820, Train Loss: 0.6419, Train Accuracy: 79.12%, Loss: 0.6379, Accuracy: 78.00%

Step: 830, Train Loss: 0.6393, Train Accuracy: 79.12%, Loss: 0.6350, Accuracy: 78.50%

Step: 840, Train Loss: 0.6365, Train Accuracy: 79.12%, Loss: 0.6320, Accuracy: 79.00%

Step: 850, Train Loss: 0.6337, Train Accuracy: 79.25%, Loss: 0.6289, Accuracy: 79.00%

Step: 860, Train Loss: 0.6307, Train Accuracy: 79.12%, Loss: 0.6257, Accuracy: 79.00%

Step: 870, Train Loss: 0.6277, Train Accuracy: 79.00%, Loss: 0.6224, Accuracy: 79.00%

Step: 880, Train Loss: 0.6245, Train Accuracy: 79.00%, Loss: 0.6190, Accuracy: 79.00%

Step: 890, Train Loss: 0.6213, Train Accuracy: 79.00%, Loss: 0.6154, Accuracy: 79.00%

Step: 900, Train Loss: 0.6180, Train Accuracy: 79.00%, Loss: 0.6118, Accuracy: 79.00%

Step: 910, Train Loss: 0.6145, Train Accuracy: 79.00%, Loss: 0.6080, Accuracy: 79.50%

Step: 920, Train Loss: 0.6110, Train Accuracy: 79.00%, Loss: 0.6042, Accuracy: 79.50%

Step: 930, Train Loss: 0.6074, Train Accuracy: 79.00%, Loss: 0.6002, Accuracy: 79.50%

Step: 940, Train Loss: 0.6038, Train Accuracy: 79.00%, Loss: 0.5962, Accuracy: 79.50%

Step: 950, Train Loss: 0.6000, Train Accuracy: 79.00%, Loss: 0.5921, Accuracy: 79.50%

Step: 960, Train Loss: 0.5962, Train Accuracy: 79.00%, Loss: 0.5879, Accuracy: 79.50%

Step: 970, Train Loss: 0.5923, Train Accuracy: 79.00%, Loss: 0.5836, Accuracy: 79.50%

Step: 980, Train Loss: 0.5884, Train Accuracy: 79.12%, Loss: 0.5792, Accuracy: 79.50%

Step: 990, Train Loss: 0.5844, Train Accuracy: 79.25%, Loss: 0.5748, Accuracy: 79.50%

Step: 1000, Train Loss: 0.5803, Train Accuracy: 79.38%, Loss: 0.5703, Accuracy: 79.50%

Step: 1010, Train Loss: 0.5762, Train Accuracy: 79.25%, Loss: 0.5657, Accuracy: 79.50%

Step: 1020, Train Loss: 0.5721, Train Accuracy: 79.25%, Loss: 0.5611, Accuracy:

79.50%

Step: 1030, Train Loss: 0.5679, Train Accuracy: 79.38%, Loss: 0.5564, Accuracy: 80.00%

Step: 1040, Train Loss: 0.5637, Train Accuracy: 79.62%, Loss: 0.5517, Accuracy: 80.50%

Step: 1050, Train Loss: 0.5594, Train Accuracy: 79.62%, Loss: 0.5470, Accuracy: 81.00%

Step: 1060, Train Loss: 0.5552, Train Accuracy: 79.62%, Loss: 0.5422, Accuracy: 81.00%

Step: 1070, Train Loss: 0.5509, Train Accuracy: 79.88%, Loss: 0.5374, Accuracy: 81.00%

Step: 1080, Train Loss: 0.5466, Train Accuracy: 79.88%, Loss: 0.5325, Accuracy: 81.50%

Step: 1090, Train Loss: 0.5423, Train Accuracy: 80.12%, Loss: 0.5277, Accuracy: 81.50%

Step: 1100, Train Loss: 0.5379, Train Accuracy: 80.00%, Loss: 0.5228, Accuracy: 82.00%

Step: 1110, Train Loss: 0.5336, Train Accuracy: 80.25%, Loss: 0.5179, Accuracy: 82.50%

Step: 1120, Train Loss: 0.5293, Train Accuracy: 80.12%, Loss: 0.5130, Accuracy: 82.50%

Step: 1130, Train Loss: 0.5250, Train Accuracy: 80.25%, Loss: 0.5081, Accuracy: 83.00%

Step: 1140, Train Loss: 0.5207, Train Accuracy: 80.50%, Loss: 0.5032, Accuracy: 83.00%

Step: 1150, Train Loss: 0.5165, Train Accuracy: 80.50%, Loss: 0.4983, Accuracy: 83.50%

Step: 1160, Train Loss: 0.5122, Train Accuracy: 80.38%, Loss: 0.4935, Accuracy: 83.50%

Step: 1170, Train Loss: 0.5080, Train Accuracy: 80.50%, Loss: 0.4886, Accuracy: 83.50%

Step: 1180, Train Loss: 0.5038, Train Accuracy: 80.38%, Loss: 0.4838, Accuracy: 83.50%

Step: 1190, Train Loss: 0.4996, Train Accuracy: 80.62%, Loss: 0.4790, Accuracy: 84.50%

Step: 1200, Train Loss: 0.4955, Train Accuracy: 80.75%, Loss: 0.4742, Accuracy: 84.50%

Step: 1210, Train Loss: 0.4914, Train Accuracy: 80.75%, Loss: 0.4695, Accuracy: 85.00%

Step: 1220, Train Loss: 0.4874, Train Accuracy: 81.00%, Loss: 0.4648, Accuracy: 85.50%

Step: 1230, Train Loss: 0.4834, Train Accuracy: 81.00%, Loss: 0.4601, Accuracy: 85.50%

Step: 1240, Train Loss: 0.4794, Train Accuracy: 81.12%, Loss: 0.4555, Accuracy: 85.50%

Step: 1250, Train Loss: 0.4755, Train Accuracy: 81.12%, Loss: 0.4509, Accuracy: 86.50%

Step: 1260, Train Loss: 0.4717, Train Accuracy: 81.25%, Loss: 0.4464, Accuracy:

87.00%

Step: 1270, Train Loss: 0.4679, Train Accuracy: 81.38%, Loss: 0.4419, Accuracy: 87.00%

Step: 1280, Train Loss: 0.4642, Train Accuracy: 81.50%, Loss: 0.4375, Accuracy: 87.00%

Step: 1290, Train Loss: 0.4605, Train Accuracy: 81.75%, Loss: 0.4331, Accuracy: 87.00%

Step: 1300, Train Loss: 0.4569, Train Accuracy: 81.75%, Loss: 0.4288, Accuracy: 87.00%

Step: 1310, Train Loss: 0.4533, Train Accuracy: 81.88%, Loss: 0.4245, Accuracy: 87.00%

Step: 1320, Train Loss: 0.4498, Train Accuracy: 81.88%, Loss: 0.4203, Accuracy: 87.00%

Step: 1330, Train Loss: 0.4464, Train Accuracy: 82.00%, Loss: 0.4162, Accuracy: 87.00%

Step: 1340, Train Loss: 0.4430, Train Accuracy: 82.00%, Loss: 0.4122, Accuracy: 87.00%

Step: 1350, Train Loss: 0.4397, Train Accuracy: 82.00%, Loss: 0.4082, Accuracy: 87.00%

Step: 1360, Train Loss: 0.4364, Train Accuracy: 82.00%, Loss: 0.4042, Accuracy: 87.00%

Step: 1370, Train Loss: 0.4332, Train Accuracy: 82.00%, Loss: 0.4004, Accuracy: 87.00%

Step: 1380, Train Loss: 0.4301, Train Accuracy: 82.00%, Loss: 0.3966, Accuracy: 87.00%

Step: 1390, Train Loss: 0.4271, Train Accuracy: 82.12%, Loss: 0.3928, Accuracy: 87.00%

Step: 1400, Train Loss: 0.4241, Train Accuracy: 82.12%, Loss: 0.3892, Accuracy: 87.00%

Step: 1410, Train Loss: 0.4211, Train Accuracy: 82.25%, Loss: 0.3856, Accuracy: 87.00%

Step: 1420, Train Loss: 0.4183, Train Accuracy: 82.50%, Loss: 0.3820, Accuracy: 87.00%

Step: 1430, Train Loss: 0.4155, Train Accuracy: 82.50%, Loss: 0.3786, Accuracy: 87.00%

Step: 1440, Train Loss: 0.4128, Train Accuracy: 82.50%, Loss: 0.3752, Accuracy: 87.00%

Step: 1450, Train Loss: 0.4101, Train Accuracy: 82.50%, Loss: 0.3719, Accuracy: 87.00%

Step: 1460, Train Loss: 0.4075, Train Accuracy: 82.50%, Loss: 0.3686, Accuracy: 87.00%

Step: 1470, Train Loss: 0.4050, Train Accuracy: 82.50%, Loss: 0.3655, Accuracy: 87.00%

Step: 1480, Train Loss: 0.4025, Train Accuracy: 82.62%, Loss: 0.3624, Accuracy: 87.00%

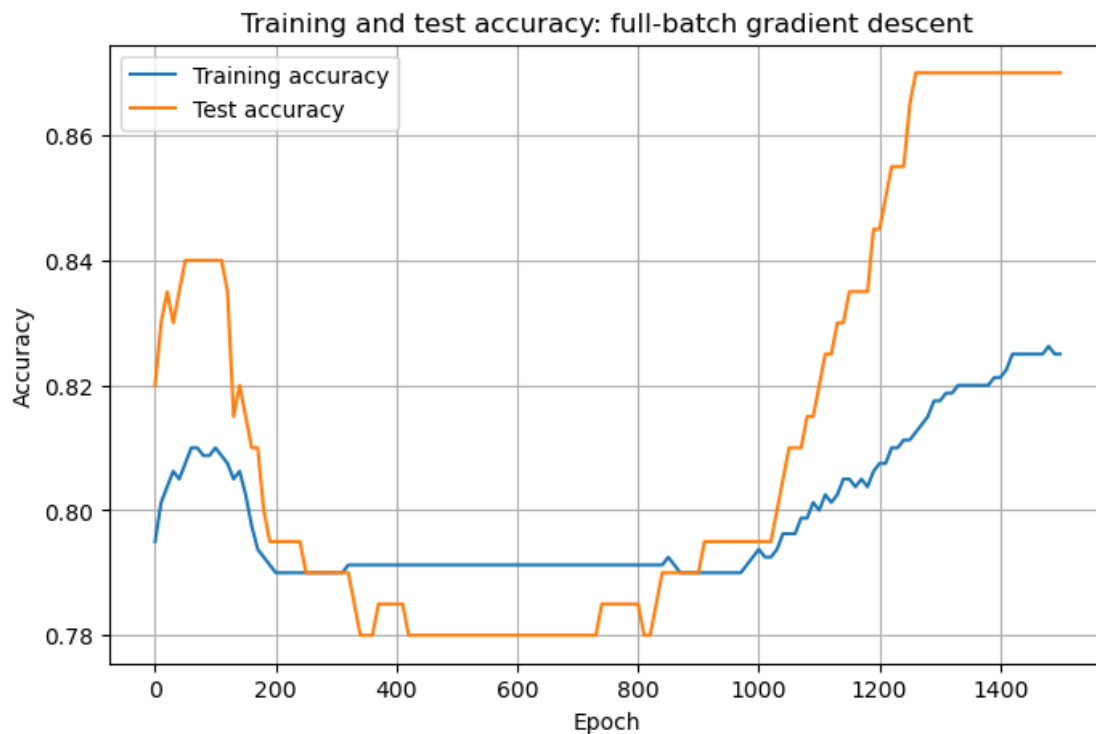
Step: 1490, Train Loss: 0.4001, Train Accuracy: 82.50%, Loss: 0.3593, Accuracy: 87.00%

Step: 1499, Train Loss: 0.3980, Train Accuracy: 82.50%, Loss: 0.3566, Accuracy:

87.00%

Training complete!

```
[3]: plt.figure(figsize=(8, 5))
plt.plot(batch_history["steps"], batch_history["train_accuracy"], label="Training accuracy")
plt.plot(batch_history["steps"], batch_history["test_accuracy"], label="Test accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.title("Training and test accuracy: full-batch gradient descent")
plt.legend()
plt.grid(True)
plt.show()
```



```
[4]: plt.figure(figsize=(8, 5))
plt.plot(batch_history["steps"], batch_history["train_loss"], label="Training loss")
plt.plot(batch_history["steps"], batch_history["test_loss"], label="Test loss")
plt.xlabel("Epoch")
plt.ylabel("Cross-entropy loss")
plt.title("Training and test loss: full-batch gradient descent")
plt.legend()
```

```
plt.grid(True)
plt.show()
```



1.2 Part III: stochastic and mini-batch gradient descent

`batch_size=1` is SGD. The other values show how mini-batch size changes the learning curves. A value of `None` means the complete training set.

```
[5]: batch_sizes = [1, 8, 16, 32, 64, 128, None]
    batch_results = {}

    for batch_size in batch_sizes:
        label = "full batch" if batch_size is None else f"batch size {batch_size}"
        _, batch_results[label] = train(
            dnn_hidden_units="20",
            learning_rate=0.01,
            max_steps=1500,
            eval_freq=10,
            batch_size=batch_size,
            random_seed=42,
        )
```

Step: 0, Train Loss: 0.6488, Train Accuracy: 64.62%, Loss: 0.6452, Accuracy: 67.00%

Step: 10, Train Loss: 0.3060, Train Accuracy: 85.12%, Loss: 0.2129, Accuracy: 92.00%

Step: 20, Train Loss: 0.3017, Train Accuracy: 85.88%, Loss: 0.2136, Accuracy: 91.00%

Step: 30, Train Loss: 0.2776, Train Accuracy: 86.50%, Loss: 0.1975, Accuracy: 92.50%

Step: 40, Train Loss: 0.2715, Train Accuracy: 86.50%, Loss: 0.1915, Accuracy: 94.50%

Step: 50, Train Loss: 0.2687, Train Accuracy: 86.75%, Loss: 0.1881, Accuracy: 95.00%

Step: 60, Train Loss: 0.2321, Train Accuracy: 89.12%, Loss: 0.1608, Accuracy: 95.50%

Step: 70, Train Loss: 0.1207, Train Accuracy: 96.50%, Loss: 0.0779, Accuracy: 98.50%

Step: 80, Train Loss: 0.0966, Train Accuracy: 96.88%, Loss: 0.0593, Accuracy: 98.50%

Step: 90, Train Loss: 0.0842, Train Accuracy: 96.75%, Loss: 0.0478, Accuracy: 99.00%

Step: 100, Train Loss: 0.0774, Train Accuracy: 97.12%, Loss: 0.0453, Accuracy: 98.50%

Step: 110, Train Loss: 0.0799, Train Accuracy: 96.88%, Loss: 0.0441, Accuracy: 98.50%

Step: 120, Train Loss: 0.0740, Train Accuracy: 97.25%, Loss: 0.0419, Accuracy: 99.00%

Step: 130, Train Loss: 0.0726, Train Accuracy: 97.25%, Loss: 0.0419, Accuracy: 99.00%

Step: 140, Train Loss: 0.0773, Train Accuracy: 97.38%, Loss: 0.0450, Accuracy: 98.50%

Step: 150, Train Loss: 0.0714, Train Accuracy: 97.62%, Loss: 0.0397, Accuracy: 98.50%

Step: 160, Train Loss: 0.0722, Train Accuracy: 97.50%, Loss: 0.0382, Accuracy: 98.50%

Step: 170, Train Loss: 0.0716, Train Accuracy: 97.12%, Loss: 0.0350, Accuracy: 99.00%

Step: 180, Train Loss: 0.0775, Train Accuracy: 97.38%, Loss: 0.0385, Accuracy: 99.00%

Step: 190, Train Loss: 0.0708, Train Accuracy: 97.50%, Loss: 0.0349, Accuracy: 99.00%

Step: 200, Train Loss: 0.0693, Train Accuracy: 97.25%, Loss: 0.0310, Accuracy: 99.00%

Step: 210, Train Loss: 0.0716, Train Accuracy: 97.25%, Loss: 0.0299, Accuracy: 99.50%

Step: 220, Train Loss: 0.0692, Train Accuracy: 97.50%, Loss: 0.0308, Accuracy: 99.50%

Step: 230, Train Loss: 0.0693, Train Accuracy: 97.50%, Loss: 0.0297, Accuracy: 99.50%

Step: 240, Train Loss: 0.0685, Train Accuracy: 97.50%, Loss: 0.0285, Accuracy: 99.50%

Step: 250, Train Loss: 0.0672, Train Accuracy: 97.50%, Loss: 0.0323, Accuracy: 99.00%

Step: 260, Train Loss: 0.0703, Train Accuracy: 97.12%, Loss: 0.0360, Accuracy: 99.50%

Step: 270, Train Loss: 0.0748, Train Accuracy: 96.88%, Loss: 0.0406, Accuracy: 98.50%

Step: 280, Train Loss: 0.0696, Train Accuracy: 97.75%, Loss: 0.0334, Accuracy: 99.50%

Step: 290, Train Loss: 0.0722, Train Accuracy: 97.50%, Loss: 0.0381, Accuracy: 99.50%

Step: 300, Train Loss: 0.0728, Train Accuracy: 96.38%, Loss: 0.0286, Accuracy: 99.50%

Step: 310, Train Loss: 0.0680, Train Accuracy: 97.75%, Loss: 0.0311, Accuracy: 99.50%

Step: 320, Train Loss: 0.0656, Train Accuracy: 97.88%, Loss: 0.0309, Accuracy: 99.00%

Step: 330, Train Loss: 0.0655, Train Accuracy: 97.88%, Loss: 0.0291, Accuracy: 99.00%

Step: 340, Train Loss: 0.0665, Train Accuracy: 97.50%, Loss: 0.0285, Accuracy: 99.00%

Step: 350, Train Loss: 0.0705, Train Accuracy: 97.00%, Loss: 0.0400, Accuracy: 99.50%

Step: 360, Train Loss: 0.0662, Train Accuracy: 97.75%, Loss: 0.0277, Accuracy: 99.50%

Step: 370, Train Loss: 0.0683, Train Accuracy: 97.75%, Loss: 0.0374, Accuracy: 99.00%

Step: 380, Train Loss: 0.0754, Train Accuracy: 96.12%, Loss: 0.0345, Accuracy: 98.50%

Step: 390, Train Loss: 0.0729, Train Accuracy: 97.00%, Loss: 0.0277, Accuracy: 99.50%

Step: 400, Train Loss: 0.0787, Train Accuracy: 95.88%, Loss: 0.0406, Accuracy: 98.50%

Step: 410, Train Loss: 0.0668, Train Accuracy: 97.62%, Loss: 0.0365, Accuracy: 99.00%

Step: 420, Train Loss: 0.0658, Train Accuracy: 97.88%, Loss: 0.0281, Accuracy: 99.50%

Step: 430, Train Loss: 0.0671, Train Accuracy: 97.88%, Loss: 0.0332, Accuracy: 99.50%

Step: 440, Train Loss: 0.0729, Train Accuracy: 97.50%, Loss: 0.0356, Accuracy: 99.00%

Step: 450, Train Loss: 0.0716, Train Accuracy: 96.88%, Loss: 0.0271, Accuracy: 99.00%

Step: 460, Train Loss: 0.0653, Train Accuracy: 97.62%, Loss: 0.0315, Accuracy: 99.00%

Step: 470, Train Loss: 0.0646, Train Accuracy: 97.62%, Loss: 0.0296, Accuracy: 99.00%

Step: 480, Train Loss: 0.0670, Train Accuracy: 97.50%, Loss: 0.0325, Accuracy: 99.00%

Step: 490, Train Loss: 0.0718, Train Accuracy: 97.38%, Loss: 0.0292, Accuracy: 99.00%

Step: 500, Train Loss: 0.0716, Train Accuracy: 97.62%, Loss: 0.0362, Accuracy: 99.00%

Step: 510, Train Loss: 0.0646, Train Accuracy: 97.88%, Loss: 0.0295, Accuracy: 99.50%

Step: 520, Train Loss: 0.0709, Train Accuracy: 97.62%, Loss: 0.0328, Accuracy: 99.00%

Step: 530, Train Loss: 0.0644, Train Accuracy: 97.88%, Loss: 0.0302, Accuracy: 99.00%

Step: 540, Train Loss: 0.0674, Train Accuracy: 97.38%, Loss: 0.0258, Accuracy: 99.50%

Step: 550, Train Loss: 0.0649, Train Accuracy: 97.62%, Loss: 0.0261, Accuracy: 99.50%

Step: 560, Train Loss: 0.0672, Train Accuracy: 97.38%, Loss: 0.0258, Accuracy: 99.50%

Step: 570, Train Loss: 0.0682, Train Accuracy: 97.38%, Loss: 0.0307, Accuracy: 99.50%

Step: 580, Train Loss: 0.0710, Train Accuracy: 96.75%, Loss: 0.0255, Accuracy: 99.00%

Step: 590, Train Loss: 0.0659, Train Accuracy: 97.75%, Loss: 0.0308, Accuracy: 99.50%

Step: 600, Train Loss: 0.0681, Train Accuracy: 97.38%, Loss: 0.0347, Accuracy: 99.50%

Step: 610, Train Loss: 0.0644, Train Accuracy: 97.88%, Loss: 0.0291, Accuracy: 99.50%

Step: 620, Train Loss: 0.0642, Train Accuracy: 98.00%, Loss: 0.0274, Accuracy: 99.50%

Step: 630, Train Loss: 0.0659, Train Accuracy: 97.75%, Loss: 0.0271, Accuracy: 99.50%

Step: 640, Train Loss: 0.0664, Train Accuracy: 97.75%, Loss: 0.0328, Accuracy: 99.50%
Step: 650, Train Loss: 0.0836, Train Accuracy: 95.50%, Loss: 0.0467, Accuracy: 97.50%
Step: 660, Train Loss: 0.0690, Train Accuracy: 97.62%, Loss: 0.0330, Accuracy: 99.00%
Step: 670, Train Loss: 0.0686, Train Accuracy: 96.88%, Loss: 0.0255, Accuracy: 99.50%
Step: 680, Train Loss: 0.0692, Train Accuracy: 97.62%, Loss: 0.0303, Accuracy: 99.00%
Step: 690, Train Loss: 0.0653, Train Accuracy: 97.62%, Loss: 0.0270, Accuracy: 99.50%
Step: 700, Train Loss: 0.0686, Train Accuracy: 97.62%, Loss: 0.0290, Accuracy: 99.00%
Step: 710, Train Loss: 0.0698, Train Accuracy: 96.62%, Loss: 0.0290, Accuracy: 99.00%
Step: 720, Train Loss: 0.0644, Train Accuracy: 98.00%, Loss: 0.0296, Accuracy: 99.00%
Step: 730, Train Loss: 0.0719, Train Accuracy: 96.50%, Loss: 0.0336, Accuracy: 99.50%
Step: 740, Train Loss: 0.0744, Train Accuracy: 97.50%, Loss: 0.0398, Accuracy: 99.00%
Step: 750, Train Loss: 0.0772, Train Accuracy: 96.88%, Loss: 0.0294, Accuracy: 99.00%
Step: 760, Train Loss: 0.0702, Train Accuracy: 97.75%, Loss: 0.0373, Accuracy: 99.50%
Step: 770, Train Loss: 0.0676, Train Accuracy: 97.38%, Loss: 0.0366, Accuracy: 99.50%
Step: 780, Train Loss: 0.0720, Train Accuracy: 96.75%, Loss: 0.0271, Accuracy: 99.00%
Step: 790, Train Loss: 0.0839, Train Accuracy: 97.12%, Loss: 0.0511, Accuracy: 97.50%
Step: 800, Train Loss: 0.0688, Train Accuracy: 97.62%, Loss: 0.0290, Accuracy: 99.00%
Step: 810, Train Loss: 0.0650, Train Accuracy: 97.75%, Loss: 0.0322, Accuracy: 99.00%
Step: 820, Train Loss: 0.0676, Train Accuracy: 97.75%, Loss: 0.0335, Accuracy: 99.50%
Step: 830, Train Loss: 0.0656, Train Accuracy: 97.62%, Loss: 0.0294, Accuracy: 99.00%
Step: 840, Train Loss: 0.0697, Train Accuracy: 97.50%, Loss: 0.0323, Accuracy: 99.00%

Step: 850, Train Loss: 0.0646, Train Accuracy: 97.88%, Loss: 0.0303, Accuracy: 99.50%

Step: 860, Train Loss: 0.0677, Train Accuracy: 97.25%, Loss: 0.0295, Accuracy: 99.00%

Step: 870, Train Loss: 0.0648, Train Accuracy: 97.75%, Loss: 0.0298, Accuracy: 99.50%

Step: 880, Train Loss: 0.0651, Train Accuracy: 97.62%, Loss: 0.0263, Accuracy: 99.50%

Step: 890, Train Loss: 0.0655, Train Accuracy: 97.62%, Loss: 0.0290, Accuracy: 99.50%

Step: 900, Train Loss: 0.0722, Train Accuracy: 97.00%, Loss: 0.0374, Accuracy: 98.50%

Step: 910, Train Loss: 0.0644, Train Accuracy: 97.88%, Loss: 0.0283, Accuracy: 99.50%

Step: 920, Train Loss: 0.0701, Train Accuracy: 97.00%, Loss: 0.0401, Accuracy: 99.50%

Step: 930, Train Loss: 0.0679, Train Accuracy: 97.62%, Loss: 0.0329, Accuracy: 99.50%

Step: 940, Train Loss: 0.0661, Train Accuracy: 97.62%, Loss: 0.0297, Accuracy: 99.00%

Step: 950, Train Loss: 0.0771, Train Accuracy: 96.62%, Loss: 0.0275, Accuracy: 99.00%

Step: 960, Train Loss: 0.0665, Train Accuracy: 97.38%, Loss: 0.0329, Accuracy: 99.00%

Step: 970, Train Loss: 0.0639, Train Accuracy: 97.75%, Loss: 0.0281, Accuracy: 99.00%

Step: 980, Train Loss: 0.0704, Train Accuracy: 96.88%, Loss: 0.0376, Accuracy: 99.50%

Step: 990, Train Loss: 0.0678, Train Accuracy: 97.75%, Loss: 0.0352, Accuracy: 99.50%

Step: 1000, Train Loss: 0.0688, Train Accuracy: 97.88%, Loss: 0.0360, Accuracy: 99.00%

Step: 1010, Train Loss: 0.0644, Train Accuracy: 98.00%, Loss: 0.0271, Accuracy: 99.00%

Step: 1020, Train Loss: 0.0689, Train Accuracy: 97.00%, Loss: 0.0266, Accuracy: 99.00%

Step: 1030, Train Loss: 0.0645, Train Accuracy: 97.88%, Loss: 0.0308, Accuracy: 99.00%

Step: 1040, Train Loss: 0.0707, Train Accuracy: 96.88%, Loss: 0.0270, Accuracy: 99.00%

Step: 1050, Train Loss: 0.0946, Train Accuracy: 94.88%, Loss: 0.0602, Accuracy: 97.00%

Step: 1060, Train Loss: 0.0695, Train Accuracy: 97.62%, Loss: 0.0337, Accuracy: 99.00%

Step: 1070, Train Loss: 0.0667, Train Accuracy: 97.62%, Loss: 0.0327, Accuracy: 99.00%

Step: 1080, Train Loss: 0.0646, Train Accuracy: 97.62%, Loss: 0.0282, Accuracy: 99.50%

Step: 1090, Train Loss: 0.0653, Train Accuracy: 97.62%, Loss: 0.0264, Accuracy: 99.50%

Step: 1100, Train Loss: 0.0707, Train Accuracy: 97.50%, Loss: 0.0311, Accuracy: 99.00%

Step: 1110, Train Loss: 0.0664, Train Accuracy: 97.50%, Loss: 0.0281, Accuracy: 99.00%

Step: 1120, Train Loss: 0.0685, Train Accuracy: 97.62%, Loss: 0.0288, Accuracy: 99.00%

Step: 1130, Train Loss: 0.0671, Train Accuracy: 97.62%, Loss: 0.0366, Accuracy: 98.50%

Step: 1140, Train Loss: 0.0670, Train Accuracy: 97.62%, Loss: 0.0285, Accuracy: 99.00%

Step: 1150, Train Loss: 0.0645, Train Accuracy: 97.75%, Loss: 0.0287, Accuracy: 99.50%

Step: 1160, Train Loss: 0.0713, Train Accuracy: 97.00%, Loss: 0.0382, Accuracy: 99.50%

Step: 1170, Train Loss: 0.0643, Train Accuracy: 97.50%, Loss: 0.0305, Accuracy: 99.00%

Step: 1180, Train Loss: 0.0652, Train Accuracy: 97.62%, Loss: 0.0266, Accuracy: 99.50%

Step: 1190, Train Loss: 0.0766, Train Accuracy: 97.25%, Loss: 0.0299, Accuracy: 98.50%

Step: 1200, Train Loss: 0.0693, Train Accuracy: 97.00%, Loss: 0.0255, Accuracy: 99.50%

Step: 1210, Train Loss: 0.0723, Train Accuracy: 97.38%, Loss: 0.0390, Accuracy: 99.00%

Step: 1220, Train Loss: 0.0682, Train Accuracy: 97.75%, Loss: 0.0359, Accuracy: 99.00%

Step: 1230, Train Loss: 0.0657, Train Accuracy: 97.62%, Loss: 0.0257, Accuracy: 99.50%

Step: 1240, Train Loss: 0.0679, Train Accuracy: 97.25%, Loss: 0.0354, Accuracy: 99.50%

Step: 1250, Train Loss: 0.0646, Train Accuracy: 97.75%, Loss: 0.0253, Accuracy: 99.50%

Step: 1260, Train Loss: 0.0869, Train Accuracy: 95.50%, Loss: 0.0496, Accuracy: 97.50%

Step: 1270, Train Loss: 0.0638, Train Accuracy: 97.88%, Loss: 0.0268, Accuracy: 99.50%

Step: 1280, Train Loss: 0.0641, Train Accuracy: 97.88%, Loss: 0.0272, Accuracy: 99.50%

Step: 1290, Train Loss: 0.0652, Train Accuracy: 97.50%, Loss: 0.0291, Accuracy: 99.50%

Step: 1300, Train Loss: 0.0648, Train Accuracy: 97.38%, Loss: 0.0304, Accuracy: 99.00%

Step: 1310, Train Loss: 0.0657, Train Accuracy: 97.88%, Loss: 0.0350, Accuracy: 98.50%

Step: 1320, Train Loss: 0.0731, Train Accuracy: 97.75%, Loss: 0.0362, Accuracy: 99.00%

Step: 1330, Train Loss: 0.0658, Train Accuracy: 97.50%, Loss: 0.0258, Accuracy: 99.50%

Step: 1340, Train Loss: 0.0663, Train Accuracy: 97.25%, Loss: 0.0251, Accuracy: 99.50%

Step: 1350, Train Loss: 0.0643, Train Accuracy: 97.50%, Loss: 0.0287, Accuracy: 99.00%

Step: 1360, Train Loss: 0.0649, Train Accuracy: 97.38%, Loss: 0.0316, Accuracy: 99.00%

Step: 1370, Train Loss: 0.0682, Train Accuracy: 96.75%, Loss: 0.0296, Accuracy: 99.00%

Step: 1380, Train Loss: 0.0677, Train Accuracy: 97.75%, Loss: 0.0273, Accuracy: 99.00%

Step: 1390, Train Loss: 0.0649, Train Accuracy: 97.50%, Loss: 0.0319, Accuracy: 99.00%

Step: 1400, Train Loss: 0.0737, Train Accuracy: 96.50%, Loss: 0.0373, Accuracy: 99.00%

Step: 1410, Train Loss: 0.0642, Train Accuracy: 97.62%, Loss: 0.0266, Accuracy: 99.50%

Step: 1420, Train Loss: 0.0638, Train Accuracy: 97.88%, Loss: 0.0270, Accuracy: 99.00%

Step: 1430, Train Loss: 0.0751, Train Accuracy: 96.38%, Loss: 0.0410, Accuracy: 98.00%

Step: 1440, Train Loss: 0.0663, Train Accuracy: 97.62%, Loss: 0.0280, Accuracy: 99.50%

Step: 1450, Train Loss: 0.0642, Train Accuracy: 98.00%, Loss: 0.0274, Accuracy: 99.00%

Step: 1460, Train Loss: 0.0658, Train Accuracy: 97.62%, Loss: 0.0269, Accuracy: 99.00%

Step: 1470, Train Loss: 0.0644, Train Accuracy: 97.88%, Loss: 0.0260, Accuracy: 99.50%

Step: 1480, Train Loss: 0.0652, Train Accuracy: 97.62%, Loss: 0.0327, Accuracy: 99.00%

Step: 1490, Train Loss: 0.0758, Train Accuracy: 96.12%, Loss: 0.0477, Accuracy: 98.00%

Step: 1499, Train Loss: 0.0745, Train Accuracy: 97.50%, Loss: 0.0433, Accuracy: 98.50%

Training complete!

Step: 0, Train Loss: 0.6926, Train Accuracy: 50.00%, Loss: 0.6926, Accuracy: 50.00%

Step: 10, Train Loss: 0.5385, Train Accuracy: 80.12%, Loss: 0.5234, Accuracy: 81.50%

Step: 20, Train Loss: 0.3269, Train Accuracy: 84.88%, Loss: 0.2589, Accuracy: 90.50%

Step: 30, Train Loss: 0.3078, Train Accuracy: 85.62%, Loss: 0.2244, Accuracy: 91.50%

Step: 40, Train Loss: 0.3054, Train Accuracy: 85.62%, Loss: 0.2170, Accuracy: 91.50%

Step: 50, Train Loss: 0.3048, Train Accuracy: 85.75%, Loss: 0.2152, Accuracy: 92.00%

Step: 60, Train Loss: 0.3046, Train Accuracy: 85.62%, Loss: 0.2148, Accuracy: 92.00%

Step: 70, Train Loss: 0.3044, Train Accuracy: 85.62%, Loss: 0.2142, Accuracy: 92.00%

Step: 80, Train Loss: 0.3043, Train Accuracy: 85.62%, Loss: 0.2137, Accuracy: 92.00%

Step: 90, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2146, Accuracy: 91.50%

Step: 100, Train Loss: 0.3040, Train Accuracy: 85.75%, Loss: 0.2137, Accuracy: 92.00%

Step: 110, Train Loss: 0.3039, Train Accuracy: 85.75%, Loss: 0.2143, Accuracy: 91.50%

Step: 120, Train Loss: 0.3036, Train Accuracy: 85.88%, Loss: 0.2141, Accuracy: 91.50%

Step: 130, Train Loss: 0.3031, Train Accuracy: 85.75%, Loss: 0.2134, Accuracy: 92.00%

Step: 140, Train Loss: 0.3028, Train Accuracy: 85.25%, Loss: 0.2123, Accuracy: 92.00%

Step: 150, Train Loss: 0.3016, Train Accuracy: 85.75%, Loss: 0.2121, Accuracy: 92.00%

Step: 160, Train Loss: 0.3004, Train Accuracy: 85.62%, Loss: 0.2114, Accuracy: 91.50%

Step: 170, Train Loss: 0.2988, Train Accuracy: 85.50%, Loss: 0.2102, Accuracy: 92.00%

Step: 180, Train Loss: 0.2971, Train Accuracy: 85.62%, Loss: 0.2086, Accuracy: 92.00%

Step: 190, Train Loss: 0.2938, Train Accuracy: 86.00%, Loss: 0.2078, Accuracy: 92.00%

Step: 200, Train Loss: 0.2906, Train Accuracy: 86.38%, Loss: 0.2059, Accuracy: 92.50%
 Step: 210, Train Loss: 0.2872, Train Accuracy: 86.62%, Loss: 0.2039, Accuracy: 92.50%
 Step: 220, Train Loss: 0.2839, Train Accuracy: 86.50%, Loss: 0.2013, Accuracy: 92.50%
 Step: 230, Train Loss: 0.2810, Train Accuracy: 86.50%, Loss: 0.1997, Accuracy: 93.00%
 Step: 240, Train Loss: 0.2786, Train Accuracy: 86.38%, Loss: 0.1975, Accuracy: 92.50%
 Step: 250, Train Loss: 0.2770, Train Accuracy: 86.62%, Loss: 0.1974, Accuracy: 93.00%
 Step: 260, Train Loss: 0.2753, Train Accuracy: 86.62%, Loss: 0.1957, Accuracy: 93.50%
 Step: 270, Train Loss: 0.2738, Train Accuracy: 86.50%, Loss: 0.1940, Accuracy: 94.50%
 Step: 280, Train Loss: 0.2729, Train Accuracy: 86.62%, Loss: 0.1931, Accuracy: 94.50%
 Step: 290, Train Loss: 0.2723, Train Accuracy: 86.38%, Loss: 0.1921, Accuracy: 94.50%
 Step: 300, Train Loss: 0.2717, Train Accuracy: 86.50%, Loss: 0.1917, Accuracy: 95.00%
 Step: 310, Train Loss: 0.2715, Train Accuracy: 86.50%, Loss: 0.1906, Accuracy: 94.50%
 Step: 320, Train Loss: 0.2712, Train Accuracy: 86.38%, Loss: 0.1902, Accuracy: 94.50%
 Step: 330, Train Loss: 0.2710, Train Accuracy: 86.62%, Loss: 0.1906, Accuracy: 95.00%
 Step: 340, Train Loss: 0.2706, Train Accuracy: 86.50%, Loss: 0.1898, Accuracy: 95.00%
 Step: 350, Train Loss: 0.2705, Train Accuracy: 86.75%, Loss: 0.1897, Accuracy: 95.00%
 Step: 360, Train Loss: 0.2704, Train Accuracy: 87.00%, Loss: 0.1894, Accuracy: 95.00%
 Step: 370, Train Loss: 0.2699, Train Accuracy: 86.75%, Loss: 0.1886, Accuracy: 95.00%
 Step: 380, Train Loss: 0.2695, Train Accuracy: 86.62%, Loss: 0.1877, Accuracy: 95.00%
 Step: 390, Train Loss: 0.2691, Train Accuracy: 86.62%, Loss: 0.1871, Accuracy: 95.00%
 Step: 400, Train Loss: 0.2687, Train Accuracy: 86.50%, Loss: 0.1864, Accuracy: 94.50%
 Step: 410, Train Loss: 0.2686, Train Accuracy: 86.88%, Loss: 0.1857, Accuracy: 94.50%
 Step: 420, Train Loss: 0.2676, Train Accuracy: 86.62%, Loss: 0.1851, Accuracy: 94.50%
 Step: 430, Train Loss: 0.2662, Train Accuracy: 86.62%, Loss: 0.1842, Accuracy:

94.50%

Step: 440, Train Loss: 0.2645, Train Accuracy: 87.00%, Loss: 0.1826, Accuracy: 94.50%

Step: 450, Train Loss: 0.2614, Train Accuracy: 87.25%, Loss: 0.1802, Accuracy: 95.00%

Step: 460, Train Loss: 0.2565, Train Accuracy: 87.75%, Loss: 0.1773, Accuracy: 95.00%

Step: 470, Train Loss: 0.2495, Train Accuracy: 88.12%, Loss: 0.1723, Accuracy: 95.50%

Step: 480, Train Loss: 0.2400, Train Accuracy: 88.75%, Loss: 0.1659, Accuracy: 95.50%

Step: 490, Train Loss: 0.2275, Train Accuracy: 89.62%, Loss: 0.1570, Accuracy: 96.00%

Step: 500, Train Loss: 0.2133, Train Accuracy: 90.38%, Loss: 0.1463, Accuracy: 96.00%

Step: 510, Train Loss: 0.1985, Train Accuracy: 91.25%, Loss: 0.1363, Accuracy: 96.00%

Step: 520, Train Loss: 0.1842, Train Accuracy: 92.00%, Loss: 0.1268, Accuracy: 96.00%

Step: 530, Train Loss: 0.1690, Train Accuracy: 93.38%, Loss: 0.1164, Accuracy: 96.00%

Step: 540, Train Loss: 0.1518, Train Accuracy: 94.75%, Loss: 0.1036, Accuracy: 97.00%

Step: 550, Train Loss: 0.1365, Train Accuracy: 95.75%, Loss: 0.0908, Accuracy: 98.00%

Step: 560, Train Loss: 0.1261, Train Accuracy: 96.25%, Loss: 0.0825, Accuracy: 98.50%

Step: 570, Train Loss: 0.1185, Train Accuracy: 96.75%, Loss: 0.0763, Accuracy: 98.50%

Step: 580, Train Loss: 0.1128, Train Accuracy: 96.62%, Loss: 0.0716, Accuracy: 98.50%

Step: 590, Train Loss: 0.1084, Train Accuracy: 96.62%, Loss: 0.0682, Accuracy: 98.50%

Step: 600, Train Loss: 0.1046, Train Accuracy: 96.62%, Loss: 0.0660, Accuracy: 99.00%

Step: 610, Train Loss: 0.1016, Train Accuracy: 96.62%, Loss: 0.0627, Accuracy: 99.00%

Step: 620, Train Loss: 0.0991, Train Accuracy: 96.75%, Loss: 0.0609, Accuracy: 99.00%

Step: 630, Train Loss: 0.0972, Train Accuracy: 96.75%, Loss: 0.0591, Accuracy: 99.00%

Step: 640, Train Loss: 0.0953, Train Accuracy: 96.75%, Loss: 0.0575, Accuracy: 99.00%

Step: 650, Train Loss: 0.0936, Train Accuracy: 96.75%, Loss: 0.0558, Accuracy: 98.50%

Step: 660, Train Loss: 0.0917, Train Accuracy: 96.62%, Loss: 0.0548, Accuracy:

98.50%

Step: 670, Train Loss: 0.0901, Train Accuracy: 96.62%, Loss: 0.0537, Accuracy: 98.50%

Step: 680, Train Loss: 0.0888, Train Accuracy: 96.75%, Loss: 0.0524, Accuracy: 98.50%

Step: 690, Train Loss: 0.0874, Train Accuracy: 96.75%, Loss: 0.0512, Accuracy: 98.50%

Step: 700, Train Loss: 0.0859, Train Accuracy: 96.75%, Loss: 0.0500, Accuracy: 98.50%

Step: 710, Train Loss: 0.0845, Train Accuracy: 96.88%, Loss: 0.0493, Accuracy: 98.50%

Step: 720, Train Loss: 0.0830, Train Accuracy: 97.12%, Loss: 0.0491, Accuracy: 98.50%

Step: 730, Train Loss: 0.0818, Train Accuracy: 97.25%, Loss: 0.0478, Accuracy: 98.50%

Step: 740, Train Loss: 0.0812, Train Accuracy: 97.12%, Loss: 0.0468, Accuracy: 98.50%

Step: 750, Train Loss: 0.0801, Train Accuracy: 97.25%, Loss: 0.0457, Accuracy: 98.50%

Step: 760, Train Loss: 0.0794, Train Accuracy: 97.38%, Loss: 0.0456, Accuracy: 98.50%

Step: 770, Train Loss: 0.0789, Train Accuracy: 97.25%, Loss: 0.0453, Accuracy: 98.50%

Step: 780, Train Loss: 0.0782, Train Accuracy: 97.25%, Loss: 0.0440, Accuracy: 98.50%

Step: 790, Train Loss: 0.0779, Train Accuracy: 97.12%, Loss: 0.0442, Accuracy: 98.50%

Step: 800, Train Loss: 0.0772, Train Accuracy: 97.25%, Loss: 0.0432, Accuracy: 98.50%

Step: 810, Train Loss: 0.0767, Train Accuracy: 97.25%, Loss: 0.0435, Accuracy: 98.50%

Step: 820, Train Loss: 0.0763, Train Accuracy: 97.25%, Loss: 0.0428, Accuracy: 98.50%

Step: 830, Train Loss: 0.0760, Train Accuracy: 97.12%, Loss: 0.0418, Accuracy: 98.50%

Step: 840, Train Loss: 0.0755, Train Accuracy: 97.25%, Loss: 0.0416, Accuracy: 98.50%

Step: 850, Train Loss: 0.0752, Train Accuracy: 97.38%, Loss: 0.0419, Accuracy: 98.50%

Step: 860, Train Loss: 0.0749, Train Accuracy: 97.38%, Loss: 0.0411, Accuracy: 98.50%

Step: 870, Train Loss: 0.0746, Train Accuracy: 97.12%, Loss: 0.0408, Accuracy: 98.50%

Step: 880, Train Loss: 0.0743, Train Accuracy: 97.50%, Loss: 0.0399, Accuracy: 98.50%

Step: 890, Train Loss: 0.0740, Train Accuracy: 97.38%, Loss: 0.0400, Accuracy: 98.50%

Step: 900, Train Loss: 0.0738, Train Accuracy: 97.50%, Loss: 0.0405, Accuracy: 98.50%
Step: 910, Train Loss: 0.0734, Train Accuracy: 97.50%, Loss: 0.0397, Accuracy: 98.50%
Step: 920, Train Loss: 0.0734, Train Accuracy: 97.38%, Loss: 0.0406, Accuracy: 98.50%
Step: 930, Train Loss: 0.0730, Train Accuracy: 97.25%, Loss: 0.0398, Accuracy: 98.50%
Step: 940, Train Loss: 0.0728, Train Accuracy: 97.38%, Loss: 0.0393, Accuracy: 98.50%
Step: 950, Train Loss: 0.0726, Train Accuracy: 97.50%, Loss: 0.0384, Accuracy: 98.50%
Step: 960, Train Loss: 0.0724, Train Accuracy: 97.38%, Loss: 0.0389, Accuracy: 98.50%
Step: 970, Train Loss: 0.0722, Train Accuracy: 97.38%, Loss: 0.0382, Accuracy: 98.50%
Step: 980, Train Loss: 0.0721, Train Accuracy: 97.50%, Loss: 0.0382, Accuracy: 98.50%

Step: 990, Train Loss: 0.0719, Train Accuracy: 97.38%, Loss: 0.0382, Accuracy: 98.50%
Step: 1000, Train Loss: 0.0717, Train Accuracy: 97.38%, Loss: 0.0384, Accuracy: 98.50%
Step: 1010, Train Loss: 0.0716, Train Accuracy: 97.38%, Loss: 0.0380, Accuracy: 98.50%
Step: 1020, Train Loss: 0.0714, Train Accuracy: 97.50%, Loss: 0.0377, Accuracy: 99.00%
Step: 1030, Train Loss: 0.0713, Train Accuracy: 97.38%, Loss: 0.0380, Accuracy: 99.00%
Step: 1040, Train Loss: 0.0711, Train Accuracy: 97.50%, Loss: 0.0370, Accuracy: 99.00%
Step: 1050, Train Loss: 0.0711, Train Accuracy: 97.38%, Loss: 0.0377, Accuracy: 99.00%
Step: 1060, Train Loss: 0.0709, Train Accuracy: 97.38%, Loss: 0.0372, Accuracy: 99.00%
Step: 1070, Train Loss: 0.0708, Train Accuracy: 97.38%, Loss: 0.0371, Accuracy: 99.00%
Step: 1080, Train Loss: 0.0707, Train Accuracy: 97.50%, Loss: 0.0373, Accuracy: 99.00%
Step: 1090, Train Loss: 0.0706, Train Accuracy: 97.38%, Loss: 0.0362, Accuracy: 99.00%

Step: 1100, Train Loss: 0.0706, Train Accuracy: 97.38%, Loss: 0.0360, Accuracy: 99.00%
Step: 1110, Train Loss: 0.0703, Train Accuracy: 97.38%, Loss: 0.0364, Accuracy: 99.00%
Step: 1120, Train Loss: 0.0702, Train Accuracy: 97.38%, Loss: 0.0360, Accuracy: 99.00%
Step: 1130, Train Loss: 0.0702, Train Accuracy: 97.50%, Loss: 0.0366, Accuracy:

99.00%

Step: 1140, Train Loss: 0.0701, Train Accuracy: 97.38%, Loss: 0.0356, Accuracy: 99.00%

Step: 1150, Train Loss: 0.0700, Train Accuracy: 97.50%, Loss: 0.0360, Accuracy: 99.00%

Step: 1160, Train Loss: 0.0699, Train Accuracy: 97.62%, Loss: 0.0360, Accuracy: 99.00%

Step: 1170, Train Loss: 0.0698, Train Accuracy: 97.50%, Loss: 0.0356, Accuracy: 99.00%

Step: 1180, Train Loss: 0.0697, Train Accuracy: 97.50%, Loss: 0.0352, Accuracy: 99.00%

Step: 1190, Train Loss: 0.0697, Train Accuracy: 97.50%, Loss: 0.0350, Accuracy: 99.00%

Step: 1200, Train Loss: 0.0696, Train Accuracy: 97.50%, Loss: 0.0350, Accuracy: 99.00%

Step: 1210, Train Loss: 0.0695, Train Accuracy: 97.50%, Loss: 0.0352, Accuracy: 99.00%

Step: 1220, Train Loss: 0.0695, Train Accuracy: 97.50%, Loss: 0.0354, Accuracy: 99.00%

Step: 1230, Train Loss: 0.0694, Train Accuracy: 97.50%, Loss: 0.0353, Accuracy: 99.00%

Step: 1240, Train Loss: 0.0694, Train Accuracy: 97.38%, Loss: 0.0354, Accuracy: 99.00%

Step: 1250, Train Loss: 0.0692, Train Accuracy: 97.62%, Loss: 0.0349, Accuracy: 99.00%

Step: 1260, Train Loss: 0.0693, Train Accuracy: 97.38%, Loss: 0.0349, Accuracy: 99.00%

Step: 1270, Train Loss: 0.0691, Train Accuracy: 97.62%, Loss: 0.0344, Accuracy: 99.00%

Step: 1280, Train Loss: 0.0690, Train Accuracy: 97.50%, Loss: 0.0347, Accuracy: 99.00%

Step: 1290, Train Loss: 0.0690, Train Accuracy: 97.62%, Loss: 0.0345, Accuracy: 99.00%

Step: 1300, Train Loss: 0.0690, Train Accuracy: 97.62%, Loss: 0.0345, Accuracy: 99.00%

Step: 1310, Train Loss: 0.0688, Train Accuracy: 97.62%, Loss: 0.0348, Accuracy: 99.00%

Step: 1320, Train Loss: 0.0693, Train Accuracy: 97.62%, Loss: 0.0344, Accuracy: 99.00%

Step: 1330, Train Loss: 0.0687, Train Accuracy: 97.62%, Loss: 0.0343, Accuracy: 99.00%

Step: 1340, Train Loss: 0.0687, Train Accuracy: 97.62%, Loss: 0.0337, Accuracy: 99.00%

Step: 1350, Train Loss: 0.0687, Train Accuracy: 97.62%, Loss: 0.0340, Accuracy: 99.00%

Step: 1360, Train Loss: 0.0687, Train Accuracy: 97.25%, Loss: 0.0349, Accuracy: 99.00%

Step: 1370, Train Loss: 0.0685, Train Accuracy: 97.50%, Loss: 0.0338, Accuracy: 99.00%
 Step: 1380, Train Loss: 0.0685, Train Accuracy: 97.62%, Loss: 0.0338, Accuracy: 99.00%
 Step: 1390, Train Loss: 0.0684, Train Accuracy: 97.62%, Loss: 0.0344, Accuracy: 99.00%
 Step: 1400, Train Loss: 0.0685, Train Accuracy: 97.38%, Loss: 0.0339, Accuracy: 99.00%
 Step: 1410, Train Loss: 0.0683, Train Accuracy: 97.62%, Loss: 0.0334, Accuracy: 99.00%
 Step: 1420, Train Loss: 0.0684, Train Accuracy: 97.62%, Loss: 0.0336, Accuracy: 99.00%

 Step: 1430, Train Loss: 0.0685, Train Accuracy: 97.38%, Loss: 0.0336, Accuracy: 99.00%
 Step: 1440, Train Loss: 0.0680, Train Accuracy: 97.62%, Loss: 0.0326, Accuracy: 99.00%
 Step: 1450, Train Loss: 0.0680, Train Accuracy: 97.62%, Loss: 0.0335, Accuracy: 99.00%
 Step: 1460, Train Loss: 0.0678, Train Accuracy: 97.62%, Loss: 0.0325, Accuracy: 99.00%
 Step: 1470, Train Loss: 0.0677, Train Accuracy: 97.62%, Loss: 0.0325, Accuracy: 99.00%
 Step: 1480, Train Loss: 0.0679, Train Accuracy: 97.62%, Loss: 0.0338, Accuracy: 99.00%
 Step: 1490, Train Loss: 0.0679, Train Accuracy: 97.62%, Loss: 0.0334, Accuracy: 99.00%
 Step: 1499, Train Loss: 0.0676, Train Accuracy: 97.62%, Loss: 0.0328, Accuracy: 99.00%
 Training complete!
 Step: 0, Train Loss: 0.6928, Train Accuracy: 50.00%, Loss: 0.6928, Accuracy: 50.00%
 Step: 10, Train Loss: 0.6825, Train Accuracy: 79.12%, Loss: 0.6817, Accuracy: 78.00%
 Step: 20, Train Loss: 0.5600, Train Accuracy: 79.62%, Loss: 0.5477, Accuracy: 81.00%
 Step: 30, Train Loss: 0.3871, Train Accuracy: 82.75%, Loss: 0.3428, Accuracy: 87.00%
 Step: 40, Train Loss: 0.3296, Train Accuracy: 84.75%, Loss: 0.2632, Accuracy: 90.00%
 Step: 50, Train Loss: 0.3131, Train Accuracy: 85.50%, Loss: 0.2355, Accuracy: 90.00%
 Step: 60, Train Loss: 0.3079, Train Accuracy: 85.38%, Loss: 0.2245, Accuracy: 91.00%

 Step: 70, Train Loss: 0.3062, Train Accuracy: 85.75%, Loss: 0.2196, Accuracy: 91.50%
 Step: 80, Train Loss: 0.3055, Train Accuracy: 85.75%, Loss: 0.2171, Accuracy: 91.50%

Step: 90, Train Loss: 0.3051, Train Accuracy: 85.62%, Loss: 0.2161, Accuracy: 92.00%
 Step: 100, Train Loss: 0.3048, Train Accuracy: 85.75%, Loss: 0.2151, Accuracy: 92.00%
 Step: 110, Train Loss: 0.3047, Train Accuracy: 85.62%, Loss: 0.2149, Accuracy: 92.00%
 Step: 120, Train Loss: 0.3046, Train Accuracy: 85.75%, Loss: 0.2146, Accuracy: 92.00%
 Step: 130, Train Loss: 0.3045, Train Accuracy: 85.62%, Loss: 0.2144, Accuracy: 92.00%
 Step: 140, Train Loss: 0.3044, Train Accuracy: 85.75%, Loss: 0.2139, Accuracy: 91.50%
 Step: 150, Train Loss: 0.3043, Train Accuracy: 85.75%, Loss: 0.2140, Accuracy: 92.00%
 Step: 160, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2139, Accuracy: 92.00%
 Step: 170, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2138, Accuracy: 91.50%
 Step: 180, Train Loss: 0.3042, Train Accuracy: 85.62%, Loss: 0.2135, Accuracy: 91.50%
 Step: 190, Train Loss: 0.3041, Train Accuracy: 85.75%, Loss: 0.2140, Accuracy: 92.00%
 Step: 200, Train Loss: 0.3040, Train Accuracy: 85.62%, Loss: 0.2138, Accuracy: 92.00%
 Step: 210, Train Loss: 0.3039, Train Accuracy: 85.62%, Loss: 0.2137, Accuracy: 92.00%
 Step: 220, Train Loss: 0.3038, Train Accuracy: 85.75%, Loss: 0.2134, Accuracy: 91.50%
 Step: 230, Train Loss: 0.3036, Train Accuracy: 85.75%, Loss: 0.2136, Accuracy: 92.00%
 Step: 240, Train Loss: 0.3035, Train Accuracy: 85.62%, Loss: 0.2134, Accuracy: 92.00%
 Step: 250, Train Loss: 0.3033, Train Accuracy: 85.88%, Loss: 0.2134, Accuracy: 92.00%
 Step: 260, Train Loss: 0.3031, Train Accuracy: 85.88%, Loss: 0.2134, Accuracy: 92.00%
 Step: 270, Train Loss: 0.3028, Train Accuracy: 85.88%, Loss: 0.2131, Accuracy: 92.00%
 Step: 280, Train Loss: 0.3025, Train Accuracy: 85.75%, Loss: 0.2129, Accuracy: 92.00%
 Step: 290, Train Loss: 0.3021, Train Accuracy: 85.75%, Loss: 0.2126, Accuracy: 92.00%
 Step: 300, Train Loss: 0.3017, Train Accuracy: 85.75%, Loss: 0.2123, Accuracy: 92.00%
 Step: 310, Train Loss: 0.3012, Train Accuracy: 85.75%, Loss: 0.2116, Accuracy: 92.00%
 Step: 320, Train Loss: 0.3005, Train Accuracy: 85.50%, Loss: 0.2113, Accuracy:

91.50%

Step: 330, Train Loss: 0.2997, Train Accuracy: 85.75%, Loss: 0.2114, Accuracy: 92.00%

Step: 340, Train Loss: 0.2988, Train Accuracy: 85.75%, Loss: 0.2106, Accuracy: 92.00%

Step: 350, Train Loss: 0.2978, Train Accuracy: 85.88%, Loss: 0.2104, Accuracy: 92.00%

Step: 360, Train Loss: 0.2966, Train Accuracy: 85.88%, Loss: 0.2096, Accuracy: 92.00%

Step: 370, Train Loss: 0.2953, Train Accuracy: 85.88%, Loss: 0.2086, Accuracy: 92.00%

Step: 380, Train Loss: 0.2938, Train Accuracy: 86.12%, Loss: 0.2075, Accuracy: 92.50%

Step: 390, Train Loss: 0.2923, Train Accuracy: 86.25%, Loss: 0.2067, Accuracy: 92.50%

Step: 400, Train Loss: 0.2907, Train Accuracy: 86.12%, Loss: 0.2054, Accuracy: 92.00%

Step: 410, Train Loss: 0.2890, Train Accuracy: 86.50%, Loss: 0.2044, Accuracy: 92.00%

Step: 420, Train Loss: 0.2873, Train Accuracy: 86.62%, Loss: 0.2034, Accuracy: 92.50%

Step: 430, Train Loss: 0.2856, Train Accuracy: 86.50%, Loss: 0.2027, Accuracy: 93.00%

Step: 440, Train Loss: 0.2841, Train Accuracy: 86.50%, Loss: 0.2014, Accuracy: 92.50%

Step: 450, Train Loss: 0.2826, Train Accuracy: 86.38%, Loss: 0.2002, Accuracy: 92.50%

Step: 460, Train Loss: 0.2811, Train Accuracy: 86.38%, Loss: 0.1994, Accuracy: 92.50%

Step: 470, Train Loss: 0.2798, Train Accuracy: 86.38%, Loss: 0.1988, Accuracy: 93.00%

Step: 480, Train Loss: 0.2787, Train Accuracy: 86.50%, Loss: 0.1982, Accuracy: 93.00%

Step: 490, Train Loss: 0.2776, Train Accuracy: 86.38%, Loss: 0.1973, Accuracy: 93.00%

Step: 500, Train Loss: 0.2766, Train Accuracy: 86.12%, Loss: 0.1962, Accuracy: 93.50%

Step: 510, Train Loss: 0.2758, Train Accuracy: 86.50%, Loss: 0.1956, Accuracy: 93.50%

Step: 520, Train Loss: 0.2751, Train Accuracy: 86.75%, Loss: 0.1951, Accuracy: 94.00%

Step: 530, Train Loss: 0.2744, Train Accuracy: 86.62%, Loss: 0.1946, Accuracy: 94.00%

Step: 540, Train Loss: 0.2738, Train Accuracy: 86.38%, Loss: 0.1939, Accuracy: 94.00%

Step: 550, Train Loss: 0.2733, Train Accuracy: 86.38%, Loss: 0.1933, Accuracy: 94.00%

Step: 560, Train Loss: 0.2730, Train Accuracy: 86.50%, Loss: 0.1932, Accuracy: 94.50%
 Step: 570, Train Loss: 0.2726, Train Accuracy: 86.38%, Loss: 0.1926, Accuracy: 94.00%
 Step: 580, Train Loss: 0.2723, Train Accuracy: 86.50%, Loss: 0.1926, Accuracy: 94.50%
 Step: 590, Train Loss: 0.2720, Train Accuracy: 86.50%, Loss: 0.1919, Accuracy: 95.00%
 Step: 600, Train Loss: 0.2718, Train Accuracy: 86.50%, Loss: 0.1919, Accuracy: 94.50%
 Step: 610, Train Loss: 0.2715, Train Accuracy: 86.25%, Loss: 0.1912, Accuracy: 94.50%
 Step: 620, Train Loss: 0.2713, Train Accuracy: 86.50%, Loss: 0.1910, Accuracy: 95.00%
 Step: 630, Train Loss: 0.2712, Train Accuracy: 86.50%, Loss: 0.1907, Accuracy: 95.00%
 Step: 640, Train Loss: 0.2710, Train Accuracy: 86.12%, Loss: 0.1904, Accuracy: 94.50%
 Step: 650, Train Loss: 0.2709, Train Accuracy: 86.38%, Loss: 0.1904, Accuracy: 95.00%
 Step: 660, Train Loss: 0.2707, Train Accuracy: 86.25%, Loss: 0.1900, Accuracy: 94.50%

 Step: 670, Train Loss: 0.2706, Train Accuracy: 86.38%, Loss: 0.1897, Accuracy: 94.50%
 Step: 680, Train Loss: 0.2706, Train Accuracy: 86.50%, Loss: 0.1893, Accuracy: 94.50%
 Step: 690, Train Loss: 0.2704, Train Accuracy: 86.38%, Loss: 0.1892, Accuracy: 94.50%
 Step: 700, Train Loss: 0.2703, Train Accuracy: 86.38%, Loss: 0.1889, Accuracy: 94.50%
 Step: 710, Train Loss: 0.2701, Train Accuracy: 86.62%, Loss: 0.1888, Accuracy: 95.00%
 Step: 720, Train Loss: 0.2701, Train Accuracy: 86.50%, Loss: 0.1883, Accuracy: 94.50%
 Step: 730, Train Loss: 0.2699, Train Accuracy: 86.62%, Loss: 0.1884, Accuracy: 95.00%
 Step: 740, Train Loss: 0.2699, Train Accuracy: 86.50%, Loss: 0.1878, Accuracy: 94.50%
 Step: 750, Train Loss: 0.2696, Train Accuracy: 86.62%, Loss: 0.1879, Accuracy: 95.00%
 Step: 760, Train Loss: 0.2694, Train Accuracy: 86.50%, Loss: 0.1875, Accuracy: 94.50%
 Step: 770, Train Loss: 0.2692, Train Accuracy: 86.50%, Loss: 0.1873, Accuracy: 94.50%
 Step: 780, Train Loss: 0.2690, Train Accuracy: 86.50%, Loss: 0.1870, Accuracy: 94.50%
 Step: 790, Train Loss: 0.2690, Train Accuracy: 86.62%, Loss: 0.1865, Accuracy:

94.50%

Step: 800, Train Loss: 0.2686, Train Accuracy: 86.38%, Loss: 0.1864, Accuracy: 94.50%

Step: 810, Train Loss: 0.2683, Train Accuracy: 86.62%, Loss: 0.1863, Accuracy: 95.00%

Step: 820, Train Loss: 0.2679, Train Accuracy: 86.75%, Loss: 0.1859, Accuracy: 95.00%

Step: 830, Train Loss: 0.2675, Train Accuracy: 86.75%, Loss: 0.1854, Accuracy: 95.00%

Step: 840, Train Loss: 0.2670, Train Accuracy: 86.75%, Loss: 0.1849, Accuracy: 94.50%

Step: 850, Train Loss: 0.2664, Train Accuracy: 86.75%, Loss: 0.1847, Accuracy: 95.00%

Step: 860, Train Loss: 0.2657, Train Accuracy: 86.75%, Loss: 0.1841, Accuracy: 95.00%

Step: 870, Train Loss: 0.2647, Train Accuracy: 87.00%, Loss: 0.1830, Accuracy: 95.00%

Step: 880, Train Loss: 0.2634, Train Accuracy: 87.38%, Loss: 0.1823, Accuracy: 95.00%

Step: 890, Train Loss: 0.2618, Train Accuracy: 87.50%, Loss: 0.1810, Accuracy: 95.00%

Step: 900, Train Loss: 0.2599, Train Accuracy: 87.50%, Loss: 0.1797, Accuracy: 95.00%

Step: 910, Train Loss: 0.2574, Train Accuracy: 87.75%, Loss: 0.1777, Accuracy: 95.00%

Step: 920, Train Loss: 0.2544, Train Accuracy: 87.75%, Loss: 0.1757, Accuracy: 95.00%

Step: 930, Train Loss: 0.2508, Train Accuracy: 88.00%, Loss: 0.1730, Accuracy: 95.50%

Step: 940, Train Loss: 0.2466, Train Accuracy: 88.38%, Loss: 0.1703, Accuracy: 95.50%

Step: 950, Train Loss: 0.2414, Train Accuracy: 89.00%, Loss: 0.1662, Accuracy: 95.50%

Step: 960, Train Loss: 0.2357, Train Accuracy: 89.38%, Loss: 0.1623, Accuracy: 95.50%

Step: 970, Train Loss: 0.2294, Train Accuracy: 89.62%, Loss: 0.1577, Accuracy: 96.00%

Step: 980, Train Loss: 0.2226, Train Accuracy: 89.88%, Loss: 0.1531, Accuracy: 96.00%

Step: 990, Train Loss: 0.2154, Train Accuracy: 90.25%, Loss: 0.1480, Accuracy: 96.00%

Step: 1000, Train Loss: 0.2081, Train Accuracy: 90.75%, Loss: 0.1432, Accuracy: 96.00%

Step: 1010, Train Loss: 0.2007, Train Accuracy: 91.12%, Loss: 0.1381, Accuracy: 96.00%

Step: 1020, Train Loss: 0.1934, Train Accuracy: 91.62%, Loss: 0.1325, Accuracy: 96.00%

Step: 1030, Train Loss: 0.1863, Train Accuracy: 91.88%, Loss: 0.1275, Accuracy: 96.00%
 Step: 1040, Train Loss: 0.1791, Train Accuracy: 92.88%, Loss: 0.1232, Accuracy: 96.00%
 Step: 1050, Train Loss: 0.1717, Train Accuracy: 93.25%, Loss: 0.1188, Accuracy: 96.00%
 Step: 1060, Train Loss: 0.1645, Train Accuracy: 93.75%, Loss: 0.1133, Accuracy: 96.00%

 Step: 1070, Train Loss: 0.1573, Train Accuracy: 94.25%, Loss: 0.1081, Accuracy: 96.50%
 Step: 1080, Train Loss: 0.1486, Train Accuracy: 95.25%, Loss: 0.1008, Accuracy: 97.00%
 Step: 1090, Train Loss: 0.1405, Train Accuracy: 95.75%, Loss: 0.0948, Accuracy: 97.50%
 Step: 1100, Train Loss: 0.1340, Train Accuracy: 96.00%, Loss: 0.0892, Accuracy: 98.00%
 Step: 1110, Train Loss: 0.1283, Train Accuracy: 96.25%, Loss: 0.0847, Accuracy: 98.50%
 Step: 1120, Train Loss: 0.1238, Train Accuracy: 96.38%, Loss: 0.0810, Accuracy: 98.50%
 Step: 1130, Train Loss: 0.1200, Train Accuracy: 96.62%, Loss: 0.0779, Accuracy: 98.50%
 Step: 1140, Train Loss: 0.1167, Train Accuracy: 96.75%, Loss: 0.0750, Accuracy: 98.50%
 Step: 1150, Train Loss: 0.1138, Train Accuracy: 96.75%, Loss: 0.0726, Accuracy: 98.50%
 Step: 1160, Train Loss: 0.1112, Train Accuracy: 96.62%, Loss: 0.0706, Accuracy: 98.50%
 Step: 1170, Train Loss: 0.1089, Train Accuracy: 96.75%, Loss: 0.0689, Accuracy: 98.50%
 Step: 1180, Train Loss: 0.1068, Train Accuracy: 96.75%, Loss: 0.0672, Accuracy: 98.50%
 Step: 1190, Train Loss: 0.1050, Train Accuracy: 96.75%, Loss: 0.0659, Accuracy: 99.00%
 Step: 1200, Train Loss: 0.1034, Train Accuracy: 96.62%, Loss: 0.0644, Accuracy: 99.00%
 Step: 1210, Train Loss: 0.1019, Train Accuracy: 96.62%, Loss: 0.0631, Accuracy: 99.00%
 Step: 1220, Train Loss: 0.1006, Train Accuracy: 96.62%, Loss: 0.0620, Accuracy: 99.00%
 Step: 1230, Train Loss: 0.0994, Train Accuracy: 96.75%, Loss: 0.0610, Accuracy: 99.00%
 Step: 1240, Train Loss: 0.0984, Train Accuracy: 96.75%, Loss: 0.0599, Accuracy: 99.00%
 Step: 1250, Train Loss: 0.0974, Train Accuracy: 96.75%, Loss: 0.0589, Accuracy: 99.00%
 Step: 1260, Train Loss: 0.0964, Train Accuracy: 96.75%, Loss: 0.0581, Accuracy:

99.00%

Step: 1270, Train Loss: 0.0955, Train Accuracy: 96.75%, Loss: 0.0573, Accuracy: 99.00%

Step: 1280, Train Loss: 0.0946, Train Accuracy: 96.75%, Loss: 0.0566, Accuracy: 99.00%

Step: 1290, Train Loss: 0.0937, Train Accuracy: 96.75%, Loss: 0.0558, Accuracy: 99.00%

Step: 1300, Train Loss: 0.0927, Train Accuracy: 96.75%, Loss: 0.0552, Accuracy: 99.00%

Step: 1310, Train Loss: 0.0918, Train Accuracy: 96.75%, Loss: 0.0548, Accuracy: 98.50%

Step: 1320, Train Loss: 0.0910, Train Accuracy: 96.75%, Loss: 0.0541, Accuracy: 99.00%

Step: 1330, Train Loss: 0.0901, Train Accuracy: 96.62%, Loss: 0.0537, Accuracy: 98.50%

Step: 1340, Train Loss: 0.0893, Train Accuracy: 96.62%, Loss: 0.0531, Accuracy: 98.50%

Step: 1350, Train Loss: 0.0887, Train Accuracy: 96.75%, Loss: 0.0528, Accuracy: 98.50%

Step: 1360, Train Loss: 0.0880, Train Accuracy: 96.75%, Loss: 0.0524, Accuracy: 98.50%

Step: 1370, Train Loss: 0.0874, Train Accuracy: 96.75%, Loss: 0.0519, Accuracy: 98.50%

Step: 1380, Train Loss: 0.0869, Train Accuracy: 96.88%, Loss: 0.0513, Accuracy: 98.50%

Step: 1390, Train Loss: 0.0863, Train Accuracy: 96.88%, Loss: 0.0509, Accuracy: 98.50%

Step: 1400, Train Loss: 0.0858, Train Accuracy: 97.00%, Loss: 0.0505, Accuracy: 98.50%

Step: 1410, Train Loss: 0.0853, Train Accuracy: 97.00%, Loss: 0.0501, Accuracy: 98.50%

Step: 1420, Train Loss: 0.0848, Train Accuracy: 97.00%, Loss: 0.0496, Accuracy: 98.50%

Step: 1430, Train Loss: 0.0844, Train Accuracy: 97.00%, Loss: 0.0494, Accuracy: 98.50%

Step: 1440, Train Loss: 0.0839, Train Accuracy: 97.00%, Loss: 0.0487, Accuracy: 98.50%

Step: 1450, Train Loss: 0.0835, Train Accuracy: 97.12%, Loss: 0.0486, Accuracy: 98.50%

Step: 1460, Train Loss: 0.0831, Train Accuracy: 97.12%, Loss: 0.0483, Accuracy: 98.50%

Step: 1470, Train Loss: 0.0826, Train Accuracy: 97.12%, Loss: 0.0478, Accuracy: 98.50%

Step: 1480, Train Loss: 0.0822, Train Accuracy: 96.88%, Loss: 0.0478, Accuracy: 98.50%

Step: 1490, Train Loss: 0.0817, Train Accuracy: 96.88%, Loss: 0.0474, Accuracy: 98.50%

Step: 1499, Train Loss: 0.0812, Train Accuracy: 97.12%, Loss: 0.0469, Accuracy: 98.50%

Training complete!

Step: 0, Train Loss: 0.6929, Train Accuracy: 59.13%, Loss: 0.6928, Accuracy: 58.00%

Step: 10, Train Loss: 0.6913, Train Accuracy: 79.00%, Loss: 0.6912, Accuracy: 79.00%

Step: 20, Train Loss: 0.6840, Train Accuracy: 79.12%, Loss: 0.6833, Accuracy: 78.00%

Step: 30, Train Loss: 0.6530, Train Accuracy: 79.12%, Loss: 0.6499, Accuracy: 78.00%

Step: 40, Train Loss: 0.5705, Train Accuracy: 79.25%, Loss: 0.5594, Accuracy: 79.50%

Step: 50, Train Loss: 0.4665, Train Accuracy: 81.50%, Loss: 0.4402, Accuracy: 87.00%

Step: 60, Train Loss: 0.3924, Train Accuracy: 82.75%, Loss: 0.3495, Accuracy: 87.00%

Step: 70, Train Loss: 0.3523, Train Accuracy: 83.50%, Loss: 0.2961, Accuracy: 88.50%

Step: 80, Train Loss: 0.3311, Train Accuracy: 84.75%, Loss: 0.2654, Accuracy: 89.50%

Step: 90, Train Loss: 0.3197, Train Accuracy: 85.50%, Loss: 0.2474, Accuracy: 90.00%

Step: 100, Train Loss: 0.3135, Train Accuracy: 85.50%, Loss: 0.2363, Accuracy: 90.00%

Step: 110, Train Loss: 0.3100, Train Accuracy: 85.62%, Loss: 0.2294, Accuracy: 90.50%

Step: 120, Train Loss: 0.3081, Train Accuracy: 85.38%, Loss: 0.2249, Accuracy: 91.00%

Step: 130, Train Loss: 0.3069, Train Accuracy: 85.62%, Loss: 0.2218, Accuracy: 91.00%

Step: 140, Train Loss: 0.3062, Train Accuracy: 85.62%, Loss: 0.2196, Accuracy: 91.50%

Step: 150, Train Loss: 0.3058, Train Accuracy: 85.88%, Loss: 0.2183, Accuracy: 91.50%

Step: 160, Train Loss: 0.3055, Train Accuracy: 85.88%, Loss: 0.2172, Accuracy: 91.50%

Step: 170, Train Loss: 0.3052, Train Accuracy: 85.75%, Loss: 0.2164, Accuracy: 91.50%

Step: 180, Train Loss: 0.3051, Train Accuracy: 85.62%, Loss: 0.2158, Accuracy: 91.50%

Step: 190, Train Loss: 0.3049, Train Accuracy: 85.75%, Loss: 0.2156, Accuracy: 92.00%

Step: 200, Train Loss: 0.3048, Train Accuracy: 85.75%, Loss: 0.2152, Accuracy: 92.00%

Step: 210, Train Loss: 0.3048, Train Accuracy: 85.75%, Loss: 0.2149, Accuracy: 92.00%

Step: 220, Train Loss: 0.3047, Train Accuracy: 85.75%, Loss: 0.2146, Accuracy:

91.50%

Step: 230, Train Loss: 0.3046, Train Accuracy: 85.62%, Loss: 0.2146, Accuracy: 92.00%

Step: 240, Train Loss: 0.3046, Train Accuracy: 85.62%, Loss: 0.2145, Accuracy: 92.00%

Step: 250, Train Loss: 0.3045, Train Accuracy: 85.75%, Loss: 0.2144, Accuracy: 92.00%

Step: 260, Train Loss: 0.3045, Train Accuracy: 85.62%, Loss: 0.2143, Accuracy: 92.00%

Step: 270, Train Loss: 0.3044, Train Accuracy: 85.62%, Loss: 0.2143, Accuracy: 92.00%

Step: 280, Train Loss: 0.3044, Train Accuracy: 85.62%, Loss: 0.2142, Accuracy: 92.00%

Step: 290, Train Loss: 0.3043, Train Accuracy: 85.75%, Loss: 0.2141, Accuracy: 92.00%

Step: 300, Train Loss: 0.3043, Train Accuracy: 85.75%, Loss: 0.2141, Accuracy: 92.00%

Step: 310, Train Loss: 0.3043, Train Accuracy: 85.75%, Loss: 0.2139, Accuracy: 91.50%

Step: 320, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2139, Accuracy: 92.00%

Step: 330, Train Loss: 0.3042, Train Accuracy: 85.62%, Loss: 0.2140, Accuracy: 92.00%

Step: 340, Train Loss: 0.3042, Train Accuracy: 85.62%, Loss: 0.2139, Accuracy: 92.00%

Step: 350, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2140, Accuracy: 92.00%

Step: 360, Train Loss: 0.3041, Train Accuracy: 85.75%, Loss: 0.2139, Accuracy: 92.00%

Step: 370, Train Loss: 0.3041, Train Accuracy: 85.62%, Loss: 0.2138, Accuracy: 92.00%

Step: 380, Train Loss: 0.3040, Train Accuracy: 85.75%, Loss: 0.2138, Accuracy: 92.00%

Step: 390, Train Loss: 0.3040, Train Accuracy: 85.62%, Loss: 0.2137, Accuracy: 92.00%

Step: 400, Train Loss: 0.3040, Train Accuracy: 85.75%, Loss: 0.2135, Accuracy: 91.50%

Step: 410, Train Loss: 0.3039, Train Accuracy: 85.75%, Loss: 0.2136, Accuracy: 91.50%

Step: 420, Train Loss: 0.3039, Train Accuracy: 85.75%, Loss: 0.2135, Accuracy: 92.00%

Step: 430, Train Loss: 0.3038, Train Accuracy: 85.62%, Loss: 0.2136, Accuracy: 92.00%

Step: 440, Train Loss: 0.3038, Train Accuracy: 85.75%, Loss: 0.2135, Accuracy: 92.00%

Step: 450, Train Loss: 0.3037, Train Accuracy: 85.75%, Loss: 0.2134, Accuracy: 91.50%

Step: 460, Train Loss: 0.3036, Train Accuracy: 85.75%, Loss: 0.2134, Accuracy: 92.00%
Step: 470, Train Loss: 0.3036, Train Accuracy: 85.62%, Loss: 0.2135, Accuracy: 92.00%
Step: 480, Train Loss: 0.3035, Train Accuracy: 85.75%, Loss: 0.2135, Accuracy: 92.00%
Step: 490, Train Loss: 0.3034, Train Accuracy: 85.62%, Loss: 0.2134, Accuracy: 92.00%
Step: 500, Train Loss: 0.3033, Train Accuracy: 85.62%, Loss: 0.2132, Accuracy: 92.00%
Step: 510, Train Loss: 0.3032, Train Accuracy: 85.62%, Loss: 0.2132, Accuracy: 92.00%
Step: 520, Train Loss: 0.3031, Train Accuracy: 85.75%, Loss: 0.2131, Accuracy: 92.00%
Step: 530, Train Loss: 0.3030, Train Accuracy: 85.75%, Loss: 0.2131, Accuracy: 92.00%
Step: 540, Train Loss: 0.3028, Train Accuracy: 85.75%, Loss: 0.2129, Accuracy: 92.00%
Step: 550, Train Loss: 0.3027, Train Accuracy: 85.75%, Loss: 0.2128, Accuracy: 92.00%
Step: 560, Train Loss: 0.3025, Train Accuracy: 85.75%, Loss: 0.2128, Accuracy: 92.00%
Step: 570, Train Loss: 0.3023, Train Accuracy: 85.75%, Loss: 0.2126, Accuracy: 92.00%
Step: 580, Train Loss: 0.3021, Train Accuracy: 85.75%, Loss: 0.2126, Accuracy: 92.00%
Step: 590, Train Loss: 0.3019, Train Accuracy: 85.75%, Loss: 0.2124, Accuracy: 92.00%
Step: 600, Train Loss: 0.3017, Train Accuracy: 85.75%, Loss: 0.2123, Accuracy: 92.00%
Step: 610, Train Loss: 0.3014, Train Accuracy: 85.75%, Loss: 0.2121, Accuracy: 92.00%
Step: 620, Train Loss: 0.3011, Train Accuracy: 85.75%, Loss: 0.2120, Accuracy: 92.00%
Step: 630, Train Loss: 0.3008, Train Accuracy: 85.75%, Loss: 0.2118, Accuracy: 92.00%
Step: 640, Train Loss: 0.3005, Train Accuracy: 85.75%, Loss: 0.2115, Accuracy: 92.00%

Step: 650, Train Loss: 0.3001, Train Accuracy: 85.75%, Loss: 0.2113, Accuracy: 92.00%
Step: 660, Train Loss: 0.2997, Train Accuracy: 85.75%, Loss: 0.2111, Accuracy: 92.00%
Step: 670, Train Loss: 0.2993, Train Accuracy: 85.75%, Loss: 0.2108, Accuracy: 92.00%
Step: 680, Train Loss: 0.2988, Train Accuracy: 85.62%, Loss: 0.2104, Accuracy: 92.00%
Step: 690, Train Loss: 0.2983, Train Accuracy: 85.75%, Loss: 0.2102, Accuracy:

92.00%

Step: 700, Train Loss: 0.2978, Train Accuracy: 85.75%, Loss: 0.2098, Accuracy: 92.00%

Step: 710, Train Loss: 0.2972, Train Accuracy: 85.75%, Loss: 0.2096, Accuracy: 92.00%

Step: 720, Train Loss: 0.2967, Train Accuracy: 85.75%, Loss: 0.2091, Accuracy: 92.00%

Step: 730, Train Loss: 0.2960, Train Accuracy: 85.75%, Loss: 0.2089, Accuracy: 92.00%

Step: 740, Train Loss: 0.2954, Train Accuracy: 85.75%, Loss: 0.2082, Accuracy: 92.00%

Step: 750, Train Loss: 0.2946, Train Accuracy: 85.88%, Loss: 0.2081, Accuracy: 92.50%

Step: 760, Train Loss: 0.2939, Train Accuracy: 86.12%, Loss: 0.2076, Accuracy: 92.50%

Step: 770, Train Loss: 0.2932, Train Accuracy: 86.00%, Loss: 0.2071, Accuracy: 92.50%

Step: 780, Train Loss: 0.2924, Train Accuracy: 86.00%, Loss: 0.2066, Accuracy: 92.50%

Step: 790, Train Loss: 0.2916, Train Accuracy: 86.00%, Loss: 0.2060, Accuracy: 92.00%

Step: 800, Train Loss: 0.2908, Train Accuracy: 86.00%, Loss: 0.2056, Accuracy: 92.00%

Step: 810, Train Loss: 0.2899, Train Accuracy: 86.00%, Loss: 0.2052, Accuracy: 92.50%

Step: 820, Train Loss: 0.2891, Train Accuracy: 86.25%, Loss: 0.2047, Accuracy: 92.50%

Step: 830, Train Loss: 0.2882, Train Accuracy: 86.62%, Loss: 0.2041, Accuracy: 92.00%

Step: 840, Train Loss: 0.2874, Train Accuracy: 86.62%, Loss: 0.2036, Accuracy: 92.00%

Step: 850, Train Loss: 0.2866, Train Accuracy: 86.50%, Loss: 0.2032, Accuracy: 92.50%

Step: 860, Train Loss: 0.2857, Train Accuracy: 86.50%, Loss: 0.2026, Accuracy: 93.00%

Step: 870, Train Loss: 0.2849, Train Accuracy: 86.62%, Loss: 0.2021, Accuracy: 92.50%

Step: 880, Train Loss: 0.2841, Train Accuracy: 86.62%, Loss: 0.2016, Accuracy: 93.00%

Step: 890, Train Loss: 0.2834, Train Accuracy: 86.50%, Loss: 0.2011, Accuracy: 93.00%

Step: 900, Train Loss: 0.2826, Train Accuracy: 86.62%, Loss: 0.2006, Accuracy: 93.00%

Step: 910, Train Loss: 0.2819, Train Accuracy: 86.50%, Loss: 0.2000, Accuracy: 92.50%

Step: 920, Train Loss: 0.2812, Train Accuracy: 86.38%, Loss: 0.1996, Accuracy: 93.00%

Step: 930, Train Loss: 0.2805, Train Accuracy: 86.38%, Loss: 0.1991, Accuracy:

93.00%

Step: 940, Train Loss: 0.2799, Train Accuracy: 86.38%, Loss: 0.1988, Accuracy: 93.00%

Step: 950, Train Loss: 0.2792, Train Accuracy: 86.25%, Loss: 0.1982, Accuracy: 92.50%

Step: 960, Train Loss: 0.2787, Train Accuracy: 86.25%, Loss: 0.1978, Accuracy: 92.50%

Step: 970, Train Loss: 0.2781, Train Accuracy: 86.50%, Loss: 0.1974, Accuracy: 92.50%

Step: 980, Train Loss: 0.2776, Train Accuracy: 86.38%, Loss: 0.1970, Accuracy: 92.50%

Step: 990, Train Loss: 0.2771, Train Accuracy: 86.38%, Loss: 0.1966, Accuracy: 93.00%

Step: 1000, Train Loss: 0.2767, Train Accuracy: 86.38%, Loss: 0.1964, Accuracy: 93.50%

Step: 1010, Train Loss: 0.2762, Train Accuracy: 86.38%, Loss: 0.1960, Accuracy: 94.00%

Step: 1020, Train Loss: 0.2758, Train Accuracy: 86.62%, Loss: 0.1955, Accuracy: 93.50%

Step: 1030, Train Loss: 0.2754, Train Accuracy: 86.62%, Loss: 0.1952, Accuracy: 93.50%

Step: 1040, Train Loss: 0.2751, Train Accuracy: 86.62%, Loss: 0.1949, Accuracy: 93.50%

Step: 1050, Train Loss: 0.2747, Train Accuracy: 86.50%, Loss: 0.1947, Accuracy: 94.00%

Step: 1060, Train Loss: 0.2744, Train Accuracy: 86.50%, Loss: 0.1943, Accuracy: 93.50%

Step: 1070, Train Loss: 0.2741, Train Accuracy: 86.50%, Loss: 0.1941, Accuracy: 94.00%

Step: 1080, Train Loss: 0.2739, Train Accuracy: 86.38%, Loss: 0.1938, Accuracy: 94.00%

Step: 1090, Train Loss: 0.2736, Train Accuracy: 86.62%, Loss: 0.1937, Accuracy: 94.50%

Step: 1100, Train Loss: 0.2734, Train Accuracy: 86.50%, Loss: 0.1934, Accuracy: 94.00%

Step: 1110, Train Loss: 0.2732, Train Accuracy: 86.50%, Loss: 0.1933, Accuracy: 94.00%

Step: 1120, Train Loss: 0.2730, Train Accuracy: 86.50%, Loss: 0.1930, Accuracy: 94.00%

Step: 1130, Train Loss: 0.2728, Train Accuracy: 86.50%, Loss: 0.1929, Accuracy: 94.00%

Step: 1140, Train Loss: 0.2726, Train Accuracy: 86.62%, Loss: 0.1928, Accuracy: 94.50%

Step: 1150, Train Loss: 0.2724, Train Accuracy: 86.50%, Loss: 0.1925, Accuracy: 94.00%

Step: 1160, Train Loss: 0.2723, Train Accuracy: 86.38%, Loss: 0.1923, Accuracy: 94.00%

Step: 1170, Train Loss: 0.2721, Train Accuracy: 86.50%, Loss: 0.1921, Accuracy: 94.50%
 Step: 1180, Train Loss: 0.2720, Train Accuracy: 86.38%, Loss: 0.1918, Accuracy: 94.50%
 Step: 1190, Train Loss: 0.2719, Train Accuracy: 86.50%, Loss: 0.1918, Accuracy: 94.50%
 Step: 1200, Train Loss: 0.2717, Train Accuracy: 86.25%, Loss: 0.1915, Accuracy: 94.50%
 Step: 1210, Train Loss: 0.2716, Train Accuracy: 86.12%, Loss: 0.1913, Accuracy: 94.50%
 Step: 1220, Train Loss: 0.2715, Train Accuracy: 86.25%, Loss: 0.1912, Accuracy: 94.50%
 Step: 1230, Train Loss: 0.2714, Train Accuracy: 86.25%, Loss: 0.1910, Accuracy: 94.50%
 Step: 1240, Train Loss: 0.2713, Train Accuracy: 86.25%, Loss: 0.1909, Accuracy: 94.50%
 Step: 1250, Train Loss: 0.2712, Train Accuracy: 86.25%, Loss: 0.1907, Accuracy: 94.50%
 Step: 1260, Train Loss: 0.2712, Train Accuracy: 86.38%, Loss: 0.1907, Accuracy: 94.50%
 Step: 1270, Train Loss: 0.2711, Train Accuracy: 86.12%, Loss: 0.1904, Accuracy: 94.50%
 Step: 1280, Train Loss: 0.2710, Train Accuracy: 86.38%, Loss: 0.1905, Accuracy: 94.50%
 Step: 1290, Train Loss: 0.2709, Train Accuracy: 86.38%, Loss: 0.1904, Accuracy: 94.50%
 Step: 1300, Train Loss: 0.2709, Train Accuracy: 86.25%, Loss: 0.1902, Accuracy: 94.50%
 Step: 1310, Train Loss: 0.2708, Train Accuracy: 86.25%, Loss: 0.1902, Accuracy: 94.50%
 Step: 1320, Train Loss: 0.2707, Train Accuracy: 86.38%, Loss: 0.1900, Accuracy: 94.50%
 Step: 1330, Train Loss: 0.2707, Train Accuracy: 86.38%, Loss: 0.1899, Accuracy: 94.50%
 Step: 1340, Train Loss: 0.2706, Train Accuracy: 86.38%, Loss: 0.1897, Accuracy: 94.50%
 Step: 1350, Train Loss: 0.2706, Train Accuracy: 86.50%, Loss: 0.1898, Accuracy: 95.00%
 Step: 1360, Train Loss: 0.2705, Train Accuracy: 86.38%, Loss: 0.1895, Accuracy: 94.50%
 Step: 1370, Train Loss: 0.2705, Train Accuracy: 86.38%, Loss: 0.1895, Accuracy: 94.50%
 Step: 1380, Train Loss: 0.2704, Train Accuracy: 86.50%, Loss: 0.1893, Accuracy: 94.50%
 Step: 1390, Train Loss: 0.2703, Train Accuracy: 86.50%, Loss: 0.1892, Accuracy: 94.50%
 Step: 1400, Train Loss: 0.2703, Train Accuracy: 86.62%, Loss: 0.1891, Accuracy: 94.50%

95.00%

Step: 1410, Train Loss: 0.2702, Train Accuracy: 86.50%, Loss: 0.1889, Accuracy: 94.50%

Step: 1420, Train Loss: 0.2702, Train Accuracy: 86.38%, Loss: 0.1887, Accuracy: 94.50%

Step: 1430, Train Loss: 0.2701, Train Accuracy: 86.38%, Loss: 0.1886, Accuracy: 94.50%

Step: 1440, Train Loss: 0.2700, Train Accuracy: 86.50%, Loss: 0.1885, Accuracy: 94.50%

Step: 1450, Train Loss: 0.2699, Train Accuracy: 86.50%, Loss: 0.1883, Accuracy: 94.50%

Step: 1460, Train Loss: 0.2699, Train Accuracy: 86.50%, Loss: 0.1883, Accuracy: 94.50%

Step: 1470, Train Loss: 0.2698, Train Accuracy: 86.50%, Loss: 0.1882, Accuracy: 94.50%

Step: 1480, Train Loss: 0.2697, Train Accuracy: 86.50%, Loss: 0.1880, Accuracy: 94.50%

Step: 1490, Train Loss: 0.2697, Train Accuracy: 86.62%, Loss: 0.1880, Accuracy: 95.00%

Step: 1499, Train Loss: 0.2696, Train Accuracy: 86.50%, Loss: 0.1878, Accuracy: 94.50%

Training complete!

Step: 0, Train Loss: 0.6929, Train Accuracy: 50.00%, Loss: 0.6929, Accuracy: 50.00%

Step: 10, Train Loss: 0.6924, Train Accuracy: 80.38%, Loss: 0.6923, Accuracy: 83.50%

Step: 20, Train Loss: 0.6913, Train Accuracy: 78.75%, Loss: 0.6912, Accuracy: 79.00%

Step: 30, Train Loss: 0.6889, Train Accuracy: 79.38%, Loss: 0.6886, Accuracy: 78.50%

Step: 40, Train Loss: 0.6835, Train Accuracy: 79.12%, Loss: 0.6828, Accuracy: 78.00%

Step: 50, Train Loss: 0.6718, Train Accuracy: 79.50%, Loss: 0.6702, Accuracy: 78.50%

Step: 60, Train Loss: 0.6489, Train Accuracy: 79.12%, Loss: 0.6454, Accuracy: 78.00%

Step: 70, Train Loss: 0.6103, Train Accuracy: 79.00%, Loss: 0.6034, Accuracy: 79.50%

Step: 80, Train Loss: 0.5588, Train Accuracy: 79.62%, Loss: 0.5462, Accuracy: 81.00%

Step: 90, Train Loss: 0.5033, Train Accuracy: 80.62%, Loss: 0.4832, Accuracy: 84.50%

Step: 100, Train Loss: 0.4528, Train Accuracy: 81.88%, Loss: 0.4240, Accuracy: 87.00%

Step: 110, Train Loss: 0.4124, Train Accuracy: 82.50%, Loss: 0.3748, Accuracy: 87.00%

Step: 120, Train Loss: 0.3826, Train Accuracy: 82.75%, Loss: 0.3369, Accuracy: 87.00%

Step: 130, Train Loss: 0.3614, Train Accuracy: 83.38%, Loss: 0.3086, Accuracy: 88.00%
 Step: 140, Train Loss: 0.3463, Train Accuracy: 83.88%, Loss: 0.2876, Accuracy: 88.50%
 Step: 150, Train Loss: 0.3352, Train Accuracy: 84.50%, Loss: 0.2717, Accuracy: 88.50%
 Step: 160, Train Loss: 0.3273, Train Accuracy: 84.88%, Loss: 0.2597, Accuracy: 90.50%
 Step: 170, Train Loss: 0.3217, Train Accuracy: 85.38%, Loss: 0.2506, Accuracy: 90.00%
 Step: 180, Train Loss: 0.3175, Train Accuracy: 85.38%, Loss: 0.2434, Accuracy: 90.00%
 Step: 190, Train Loss: 0.3144, Train Accuracy: 85.62%, Loss: 0.2381, Accuracy: 90.00%
 Step: 200, Train Loss: 0.3121, Train Accuracy: 85.50%, Loss: 0.2335, Accuracy: 90.50%

 Step: 210, Train Loss: 0.3104, Train Accuracy: 85.62%, Loss: 0.2301, Accuracy: 90.50%
 Step: 220, Train Loss: 0.3091, Train Accuracy: 85.62%, Loss: 0.2273, Accuracy: 91.00%
 Step: 230, Train Loss: 0.3082, Train Accuracy: 85.50%, Loss: 0.2251, Accuracy: 91.00%
 Step: 240, Train Loss: 0.3075, Train Accuracy: 85.38%, Loss: 0.2233, Accuracy: 91.00%
 Step: 250, Train Loss: 0.3070, Train Accuracy: 85.38%, Loss: 0.2218, Accuracy: 91.00%
 Step: 260, Train Loss: 0.3065, Train Accuracy: 85.62%, Loss: 0.2207, Accuracy: 91.00%
 Step: 270, Train Loss: 0.3062, Train Accuracy: 85.75%, Loss: 0.2196, Accuracy: 91.50%
 Step: 280, Train Loss: 0.3060, Train Accuracy: 85.88%, Loss: 0.2188, Accuracy: 91.50%
 Step: 290, Train Loss: 0.3058, Train Accuracy: 85.88%, Loss: 0.2182, Accuracy: 91.50%
 Step: 300, Train Loss: 0.3056, Train Accuracy: 85.88%, Loss: 0.2176, Accuracy: 91.50%
 Step: 310, Train Loss: 0.3054, Train Accuracy: 85.62%, Loss: 0.2170, Accuracy: 91.50%
 Step: 320, Train Loss: 0.3053, Train Accuracy: 85.75%, Loss: 0.2167, Accuracy: 92.00%
 Step: 330, Train Loss: 0.3052, Train Accuracy: 85.75%, Loss: 0.2163, Accuracy: 92.00%
 Step: 340, Train Loss: 0.3051, Train Accuracy: 85.75%, Loss: 0.2161, Accuracy: 92.00%
 Step: 350, Train Loss: 0.3050, Train Accuracy: 85.75%, Loss: 0.2159, Accuracy: 92.00%
 Step: 360, Train Loss: 0.3050, Train Accuracy: 85.75%, Loss: 0.2157, Accuracy:

92.00%

Step: 370, Train Loss: 0.3049, Train Accuracy: 85.75%, Loss: 0.2154, Accuracy: 92.00%

Step: 380, Train Loss: 0.3049, Train Accuracy: 85.75%, Loss: 0.2153, Accuracy: 92.00%

Step: 390, Train Loss: 0.3048, Train Accuracy: 85.75%, Loss: 0.2151, Accuracy: 92.00%

Step: 400, Train Loss: 0.3048, Train Accuracy: 85.75%, Loss: 0.2149, Accuracy: 91.50%

Step: 410, Train Loss: 0.3047, Train Accuracy: 85.75%, Loss: 0.2148, Accuracy: 92.00%

Step: 420, Train Loss: 0.3047, Train Accuracy: 85.75%, Loss: 0.2147, Accuracy: 91.50%

Step: 430, Train Loss: 0.3047, Train Accuracy: 85.62%, Loss: 0.2149, Accuracy: 92.00%

Step: 440, Train Loss: 0.3046, Train Accuracy: 85.75%, Loss: 0.2147, Accuracy: 92.00%

Step: 450, Train Loss: 0.3046, Train Accuracy: 85.62%, Loss: 0.2146, Accuracy: 92.00%

Step: 460, Train Loss: 0.3046, Train Accuracy: 85.75%, Loss: 0.2144, Accuracy: 92.00%

Step: 470, Train Loss: 0.3045, Train Accuracy: 85.75%, Loss: 0.2143, Accuracy: 91.50%

Step: 480, Train Loss: 0.3045, Train Accuracy: 85.75%, Loss: 0.2145, Accuracy: 92.00%

Step: 490, Train Loss: 0.3045, Train Accuracy: 85.62%, Loss: 0.2143, Accuracy: 92.00%

Step: 500, Train Loss: 0.3045, Train Accuracy: 85.62%, Loss: 0.2142, Accuracy: 92.00%

Step: 510, Train Loss: 0.3044, Train Accuracy: 85.75%, Loss: 0.2142, Accuracy: 92.00%

Step: 520, Train Loss: 0.3044, Train Accuracy: 85.75%, Loss: 0.2141, Accuracy: 92.00%

Step: 530, Train Loss: 0.3044, Train Accuracy: 85.75%, Loss: 0.2143, Accuracy: 92.00%

Step: 540, Train Loss: 0.3044, Train Accuracy: 85.75%, Loss: 0.2140, Accuracy: 91.50%

Step: 550, Train Loss: 0.3044, Train Accuracy: 85.75%, Loss: 0.2140, Accuracy: 92.00%

Step: 560, Train Loss: 0.3044, Train Accuracy: 85.75%, Loss: 0.2141, Accuracy: 92.00%

Step: 570, Train Loss: 0.3043, Train Accuracy: 85.75%, Loss: 0.2140, Accuracy: 92.00%

Step: 580, Train Loss: 0.3043, Train Accuracy: 85.75%, Loss: 0.2141, Accuracy: 92.00%

Step: 590, Train Loss: 0.3043, Train Accuracy: 85.62%, Loss: 0.2139, Accuracy: 92.00%

Step: 600, Train Loss: 0.3043, Train Accuracy: 85.62%, Loss: 0.2140, Accuracy:

92.00%

Step: 610, Train Loss: 0.3043, Train Accuracy: 85.75%, Loss: 0.2139, Accuracy: 92.00%

Step: 620, Train Loss: 0.3042, Train Accuracy: 85.62%, Loss: 0.2138, Accuracy: 92.00%

Step: 630, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2136, Accuracy: 92.00%

Step: 640, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2136, Accuracy: 91.50%

Step: 650, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2138, Accuracy: 92.00%

Step: 660, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2138, Accuracy: 92.00%

Step: 670, Train Loss: 0.3041, Train Accuracy: 85.75%, Loss: 0.2137, Accuracy: 92.00%

Step: 680, Train Loss: 0.3041, Train Accuracy: 85.62%, Loss: 0.2138, Accuracy: 92.00%

Step: 690, Train Loss: 0.3041, Train Accuracy: 85.62%, Loss: 0.2138, Accuracy: 92.00%

Step: 700, Train Loss: 0.3041, Train Accuracy: 85.75%, Loss: 0.2137, Accuracy: 92.00%

Step: 710, Train Loss: 0.3041, Train Accuracy: 85.75%, Loss: 0.2139, Accuracy: 92.00%

Step: 720, Train Loss: 0.3041, Train Accuracy: 85.88%, Loss: 0.2135, Accuracy: 91.50%

Step: 730, Train Loss: 0.3040, Train Accuracy: 85.62%, Loss: 0.2137, Accuracy: 92.00%

Step: 740, Train Loss: 0.3040, Train Accuracy: 85.75%, Loss: 0.2135, Accuracy: 91.50%

Step: 750, Train Loss: 0.3040, Train Accuracy: 85.62%, Loss: 0.2139, Accuracy: 92.00%

Step: 760, Train Loss: 0.3040, Train Accuracy: 85.75%, Loss: 0.2136, Accuracy: 92.00%

Step: 770, Train Loss: 0.3039, Train Accuracy: 85.75%, Loss: 0.2137, Accuracy: 92.00%

Step: 780, Train Loss: 0.3039, Train Accuracy: 85.75%, Loss: 0.2136, Accuracy: 92.00%

Step: 790, Train Loss: 0.3039, Train Accuracy: 85.62%, Loss: 0.2135, Accuracy: 91.50%

Step: 800, Train Loss: 0.3039, Train Accuracy: 85.75%, Loss: 0.2135, Accuracy: 91.50%

Step: 810, Train Loss: 0.3039, Train Accuracy: 85.62%, Loss: 0.2136, Accuracy: 92.00%

Step: 820, Train Loss: 0.3038, Train Accuracy: 85.62%, Loss: 0.2136, Accuracy: 92.00%

Step: 830, Train Loss: 0.3038, Train Accuracy: 85.62%, Loss: 0.2136, Accuracy: 92.00%

Step: 840, Train Loss: 0.3038, Train Accuracy: 85.62%, Loss: 0.2136, Accuracy: 92.00%
Step: 850, Train Loss: 0.3037, Train Accuracy: 85.62%, Loss: 0.2135, Accuracy: 92.00%
Step: 860, Train Loss: 0.3037, Train Accuracy: 85.62%, Loss: 0.2135, Accuracy: 92.00%
Step: 870, Train Loss: 0.3037, Train Accuracy: 85.75%, Loss: 0.2133, Accuracy: 92.00%
Step: 880, Train Loss: 0.3037, Train Accuracy: 85.75%, Loss: 0.2137, Accuracy: 92.00%
Step: 890, Train Loss: 0.3036, Train Accuracy: 85.62%, Loss: 0.2134, Accuracy: 92.00%
Step: 900, Train Loss: 0.3036, Train Accuracy: 85.75%, Loss: 0.2136, Accuracy: 92.00%
Step: 910, Train Loss: 0.3035, Train Accuracy: 85.62%, Loss: 0.2134, Accuracy: 92.00%
Step: 920, Train Loss: 0.3035, Train Accuracy: 85.62%, Loss: 0.2134, Accuracy: 92.00%
Step: 930, Train Loss: 0.3035, Train Accuracy: 85.62%, Loss: 0.2134, Accuracy: 92.00%
Step: 940, Train Loss: 0.3034, Train Accuracy: 85.75%, Loss: 0.2133, Accuracy: 92.00%
Step: 950, Train Loss: 0.3034, Train Accuracy: 85.88%, Loss: 0.2133, Accuracy: 92.00%
Step: 960, Train Loss: 0.3033, Train Accuracy: 85.62%, Loss: 0.2131, Accuracy: 92.00%
Step: 970, Train Loss: 0.3033, Train Accuracy: 85.75%, Loss: 0.2132, Accuracy: 92.00%
Step: 980, Train Loss: 0.3032, Train Accuracy: 85.62%, Loss: 0.2131, Accuracy: 92.00%
Step: 990, Train Loss: 0.3032, Train Accuracy: 85.62%, Loss: 0.2130, Accuracy: 91.50%
Step: 1000, Train Loss: 0.3031, Train Accuracy: 85.75%, Loss: 0.2132, Accuracy: 92.00%
Step: 1010, Train Loss: 0.3030, Train Accuracy: 85.75%, Loss: 0.2131, Accuracy: 92.00%
Step: 1020, Train Loss: 0.3030, Train Accuracy: 85.75%, Loss: 0.2131, Accuracy: 92.00%
Step: 1030, Train Loss: 0.3029, Train Accuracy: 85.88%, Loss: 0.2128, Accuracy: 91.50%
Step: 1040, Train Loss: 0.3028, Train Accuracy: 85.75%, Loss: 0.2128, Accuracy: 92.00%
Step: 1050, Train Loss: 0.3028, Train Accuracy: 85.88%, Loss: 0.2130, Accuracy: 92.00%
Step: 1060, Train Loss: 0.3027, Train Accuracy: 85.75%, Loss: 0.2128, Accuracy: 92.00%
Step: 1070, Train Loss: 0.3026, Train Accuracy: 85.75%, Loss: 0.2128, Accuracy: 92.00%

Step: 1080, Train Loss: 0.3025, Train Accuracy: 85.88%, Loss: 0.2126, Accuracy: 91.50%
 Step: 1090, Train Loss: 0.3024, Train Accuracy: 85.75%, Loss: 0.2126, Accuracy: 92.00%
 Step: 1100, Train Loss: 0.3023, Train Accuracy: 85.75%, Loss: 0.2126, Accuracy: 92.00%
 Step: 1110, Train Loss: 0.3022, Train Accuracy: 85.75%, Loss: 0.2126, Accuracy: 92.00%
 Step: 1120, Train Loss: 0.3021, Train Accuracy: 85.75%, Loss: 0.2125, Accuracy: 92.00%
 Step: 1130, Train Loss: 0.3020, Train Accuracy: 85.75%, Loss: 0.2124, Accuracy: 92.00%
 Step: 1140, Train Loss: 0.3019, Train Accuracy: 85.75%, Loss: 0.2122, Accuracy: 92.00%
 Step: 1150, Train Loss: 0.3017, Train Accuracy: 85.75%, Loss: 0.2123, Accuracy: 92.00%
 Step: 1160, Train Loss: 0.3016, Train Accuracy: 85.75%, Loss: 0.2121, Accuracy: 92.00%
 Step: 1170, Train Loss: 0.3015, Train Accuracy: 85.75%, Loss: 0.2122, Accuracy: 92.00%
 Step: 1180, Train Loss: 0.3013, Train Accuracy: 85.75%, Loss: 0.2119, Accuracy: 91.50%
 Step: 1190, Train Loss: 0.3012, Train Accuracy: 85.75%, Loss: 0.2119, Accuracy: 92.00%
 Step: 1200, Train Loss: 0.3010, Train Accuracy: 85.75%, Loss: 0.2119, Accuracy: 92.00%
 Step: 1210, Train Loss: 0.3009, Train Accuracy: 85.75%, Loss: 0.2117, Accuracy: 92.00%
 Step: 1220, Train Loss: 0.3007, Train Accuracy: 85.75%, Loss: 0.2116, Accuracy: 92.00%
 Step: 1230, Train Loss: 0.3005, Train Accuracy: 85.75%, Loss: 0.2115, Accuracy: 92.00%
 Step: 1240, Train Loss: 0.3003, Train Accuracy: 85.62%, Loss: 0.2112, Accuracy: 91.50%
 Step: 1250, Train Loss: 0.3002, Train Accuracy: 85.75%, Loss: 0.2112, Accuracy: 92.00%
 Step: 1260, Train Loss: 0.3000, Train Accuracy: 85.75%, Loss: 0.2114, Accuracy: 92.00%
 Step: 1270, Train Loss: 0.2998, Train Accuracy: 85.50%, Loss: 0.2108, Accuracy: 92.00%
 Step: 1280, Train Loss: 0.2995, Train Accuracy: 85.75%, Loss: 0.2109, Accuracy: 92.00%
 Step: 1290, Train Loss: 0.2993, Train Accuracy: 85.75%, Loss: 0.2107, Accuracy: 92.00%
 Step: 1300, Train Loss: 0.2991, Train Accuracy: 85.75%, Loss: 0.2107, Accuracy: 92.00%
 Step: 1310, Train Loss: 0.2988, Train Accuracy: 85.75%, Loss: 0.2104, Accuracy: 92.00%

Step: 1320, Train Loss: 0.2986, Train Accuracy: 85.88%, Loss: 0.2103, Accuracy: 92.00%
 Step: 1330, Train Loss: 0.2983, Train Accuracy: 85.75%, Loss: 0.2103, Accuracy: 92.00%
 Step: 1340, Train Loss: 0.2980, Train Accuracy: 85.75%, Loss: 0.2099, Accuracy: 91.50%
 Step: 1350, Train Loss: 0.2977, Train Accuracy: 85.62%, Loss: 0.2100, Accuracy: 92.00%
 Step: 1360, Train Loss: 0.2974, Train Accuracy: 85.88%, Loss: 0.2097, Accuracy: 92.50%
 Step: 1370, Train Loss: 0.2972, Train Accuracy: 85.75%, Loss: 0.2097, Accuracy: 92.00%
 Step: 1380, Train Loss: 0.2968, Train Accuracy: 85.75%, Loss: 0.2094, Accuracy: 92.00%
 Step: 1390, Train Loss: 0.2965, Train Accuracy: 85.75%, Loss: 0.2092, Accuracy: 92.50%
 Step: 1400, Train Loss: 0.2962, Train Accuracy: 85.75%, Loss: 0.2090, Accuracy: 92.00%
 Step: 1410, Train Loss: 0.2959, Train Accuracy: 85.62%, Loss: 0.2087, Accuracy: 92.00%
 Step: 1420, Train Loss: 0.2955, Train Accuracy: 85.75%, Loss: 0.2086, Accuracy: 92.00%
 Step: 1430, Train Loss: 0.2952, Train Accuracy: 85.88%, Loss: 0.2084, Accuracy: 92.50%
 Step: 1440, Train Loss: 0.2948, Train Accuracy: 85.88%, Loss: 0.2080, Accuracy: 92.00%
 Step: 1450, Train Loss: 0.2944, Train Accuracy: 85.88%, Loss: 0.2078, Accuracy: 92.00%
 Step: 1460, Train Loss: 0.2940, Train Accuracy: 86.00%, Loss: 0.2077, Accuracy: 92.50%
 Step: 1470, Train Loss: 0.2937, Train Accuracy: 86.12%, Loss: 0.2076, Accuracy: 92.50%
 Step: 1480, Train Loss: 0.2933, Train Accuracy: 85.88%, Loss: 0.2070, Accuracy: 92.00%
 Step: 1490, Train Loss: 0.2928, Train Accuracy: 86.12%, Loss: 0.2071, Accuracy: 92.50%
 Step: 1499, Train Loss: 0.2925, Train Accuracy: 86.12%, Loss: 0.2068, Accuracy: 92.50%
 Training complete!
 Step: 0, Train Loss: 0.6929, Train Accuracy: 50.00%, Loss: 0.6929, Accuracy: 50.00%
 Step: 10, Train Loss: 0.6927, Train Accuracy: 80.88%, Loss: 0.6927, Accuracy: 84.00%
 Step: 20, Train Loss: 0.6924, Train Accuracy: 80.88%, Loss: 0.6923, Accuracy: 85.00%
 Step: 30, Train Loss: 0.6919, Train Accuracy: 79.12%, Loss: 0.6918, Accuracy: 79.50%

Step: 40, Train Loss: 0.6912, Train Accuracy: 79.12%, Loss: 0.6910, Accuracy: 79.00%

Step: 50, Train Loss: 0.6900, Train Accuracy: 78.88%, Loss: 0.6898, Accuracy: 78.00%

Step: 60, Train Loss: 0.6882, Train Accuracy: 79.00%, Loss: 0.6879, Accuracy: 79.00%

Step: 70, Train Loss: 0.6855, Train Accuracy: 78.38%, Loss: 0.6849, Accuracy: 82.50%

Step: 80, Train Loss: 0.6813, Train Accuracy: 78.62%, Loss: 0.6804, Accuracy: 79.00%

Step: 90, Train Loss: 0.6750, Train Accuracy: 78.50%, Loss: 0.6737, Accuracy: 78.50%

Step: 100, Train Loss: 0.6658, Train Accuracy: 78.50%, Loss: 0.6637, Accuracy: 78.50%

Step: 110, Train Loss: 0.6526, Train Accuracy: 79.00%, Loss: 0.6495, Accuracy: 79.00%

Step: 120, Train Loss: 0.6350, Train Accuracy: 79.00%, Loss: 0.6304, Accuracy: 79.50%

Step: 130, Train Loss: 0.6126, Train Accuracy: 79.00%, Loss: 0.6059, Accuracy: 79.50%

Step: 140, Train Loss: 0.5862, Train Accuracy: 79.00%, Loss: 0.5768, Accuracy: 79.50%

Step: 150, Train Loss: 0.5569, Train Accuracy: 79.50%, Loss: 0.5442, Accuracy: 81.00%

Step: 160, Train Loss: 0.5270, Train Accuracy: 80.38%, Loss: 0.5104, Accuracy: 83.00%

Step: 170, Train Loss: 0.4978, Train Accuracy: 80.75%, Loss: 0.4769, Accuracy: 85.00%

Step: 180, Train Loss: 0.4700, Train Accuracy: 81.12%, Loss: 0.4443, Accuracy: 86.50%

Step: 190, Train Loss: 0.4449, Train Accuracy: 82.00%, Loss: 0.4144, Accuracy: 87.00%

Step: 200, Train Loss: 0.4227, Train Accuracy: 82.12%, Loss: 0.3874, Accuracy: 87.00%

Step: 210, Train Loss: 0.4037, Train Accuracy: 82.50%, Loss: 0.3639, Accuracy: 87.00%

Step: 220, Train Loss: 0.3878, Train Accuracy: 82.62%, Loss: 0.3436, Accuracy: 87.00%

Step: 230, Train Loss: 0.3745, Train Accuracy: 82.75%, Loss: 0.3262, Accuracy: 87.50%

Step: 240, Train Loss: 0.3633, Train Accuracy: 83.25%, Loss: 0.3113, Accuracy: 88.00%

Step: 250, Train Loss: 0.3542, Train Accuracy: 83.38%, Loss: 0.2988, Accuracy: 88.50%

Step: 260, Train Loss: 0.3466, Train Accuracy: 83.75%, Loss: 0.2882, Accuracy: 88.50%

Step: 270, Train Loss: 0.3402, Train Accuracy: 84.00%, Loss: 0.2790, Accuracy: 88.50%

Step: 280, Train Loss: 0.3348, Train Accuracy: 84.62%, Loss: 0.2710, Accuracy: 88.50%
Step: 290, Train Loss: 0.3303, Train Accuracy: 84.75%, Loss: 0.2642, Accuracy: 89.50%
Step: 300, Train Loss: 0.3266, Train Accuracy: 84.88%, Loss: 0.2583, Accuracy: 90.50%
Step: 310, Train Loss: 0.3233, Train Accuracy: 85.25%, Loss: 0.2531, Accuracy: 90.50%
Step: 320, Train Loss: 0.3206, Train Accuracy: 85.38%, Loss: 0.2489, Accuracy: 90.00%
Step: 330, Train Loss: 0.3183, Train Accuracy: 85.75%, Loss: 0.2449, Accuracy: 90.00%
Step: 340, Train Loss: 0.3164, Train Accuracy: 85.50%, Loss: 0.2417, Accuracy: 90.00%
Step: 350, Train Loss: 0.3149, Train Accuracy: 85.50%, Loss: 0.2389, Accuracy: 90.50%
Step: 360, Train Loss: 0.3135, Train Accuracy: 85.50%, Loss: 0.2363, Accuracy: 90.00%
Step: 370, Train Loss: 0.3123, Train Accuracy: 85.62%, Loss: 0.2340, Accuracy: 90.50%
Step: 380, Train Loss: 0.3113, Train Accuracy: 85.75%, Loss: 0.2320, Accuracy: 90.50%
Step: 390, Train Loss: 0.3105, Train Accuracy: 85.50%, Loss: 0.2302, Accuracy: 90.50%
Step: 400, Train Loss: 0.3098, Train Accuracy: 85.50%, Loss: 0.2287, Accuracy: 90.50%
Step: 410, Train Loss: 0.3091, Train Accuracy: 85.62%, Loss: 0.2273, Accuracy: 91.00%
Step: 420, Train Loss: 0.3086, Train Accuracy: 85.75%, Loss: 0.2260, Accuracy: 91.00%
Step: 430, Train Loss: 0.3082, Train Accuracy: 85.50%, Loss: 0.2253, Accuracy: 91.00%
Step: 440, Train Loss: 0.3078, Train Accuracy: 85.38%, Loss: 0.2243, Accuracy: 91.00%
Step: 450, Train Loss: 0.3075, Train Accuracy: 85.62%, Loss: 0.2235, Accuracy: 91.00%
Step: 460, Train Loss: 0.3071, Train Accuracy: 85.38%, Loss: 0.2223, Accuracy: 91.00%
Step: 470, Train Loss: 0.3069, Train Accuracy: 85.62%, Loss: 0.2214, Accuracy: 91.50%
Step: 480, Train Loss: 0.3066, Train Accuracy: 85.62%, Loss: 0.2209, Accuracy: 91.00%
Step: 490, Train Loss: 0.3064, Train Accuracy: 85.75%, Loss: 0.2203, Accuracy: 91.50%
Step: 500, Train Loss: 0.3063, Train Accuracy: 85.75%, Loss: 0.2199, Accuracy: 91.50%
Step: 510, Train Loss: 0.3061, Train Accuracy: 85.75%, Loss: 0.2194, Accuracy: 91.50%

Step: 520, Train Loss: 0.3060, Train Accuracy: 85.75%, Loss: 0.2189, Accuracy: 91.50%
 Step: 530, Train Loss: 0.3059, Train Accuracy: 85.75%, Loss: 0.2188, Accuracy: 91.50%
 Step: 540, Train Loss: 0.3058, Train Accuracy: 85.75%, Loss: 0.2180, Accuracy: 91.50%
 Step: 550, Train Loss: 0.3056, Train Accuracy: 85.88%, Loss: 0.2177, Accuracy: 91.50%
 Step: 560, Train Loss: 0.3056, Train Accuracy: 85.75%, Loss: 0.2176, Accuracy: 91.50%
 Step: 570, Train Loss: 0.3055, Train Accuracy: 85.88%, Loss: 0.2172, Accuracy: 92.00%
 Step: 580, Train Loss: 0.3054, Train Accuracy: 85.88%, Loss: 0.2170, Accuracy: 91.50%
 Step: 590, Train Loss: 0.3053, Train Accuracy: 85.75%, Loss: 0.2166, Accuracy: 91.50%
 Step: 600, Train Loss: 0.3053, Train Accuracy: 85.75%, Loss: 0.2166, Accuracy: 92.00%
 Step: 610, Train Loss: 0.3052, Train Accuracy: 85.62%, Loss: 0.2165, Accuracy: 92.00%
 Step: 620, Train Loss: 0.3052, Train Accuracy: 85.75%, Loss: 0.2160, Accuracy: 92.00%
 Step: 630, Train Loss: 0.3051, Train Accuracy: 85.62%, Loss: 0.2156, Accuracy: 91.50%
 Step: 640, Train Loss: 0.3051, Train Accuracy: 85.62%, Loss: 0.2154, Accuracy: 91.50%
 Step: 650, Train Loss: 0.3050, Train Accuracy: 85.75%, Loss: 0.2156, Accuracy: 92.00%
 Step: 660, Train Loss: 0.3050, Train Accuracy: 85.75%, Loss: 0.2155, Accuracy: 92.00%
 Step: 670, Train Loss: 0.3050, Train Accuracy: 85.75%, Loss: 0.2153, Accuracy: 92.00%
 Step: 680, Train Loss: 0.3049, Train Accuracy: 85.62%, Loss: 0.2154, Accuracy: 92.00%
 Step: 690, Train Loss: 0.3049, Train Accuracy: 85.62%, Loss: 0.2153, Accuracy: 92.00%
 Step: 700, Train Loss: 0.3049, Train Accuracy: 85.75%, Loss: 0.2151, Accuracy: 92.00%
 Step: 710, Train Loss: 0.3049, Train Accuracy: 85.75%, Loss: 0.2153, Accuracy: 92.00%
 Step: 720, Train Loss: 0.3048, Train Accuracy: 85.75%, Loss: 0.2147, Accuracy: 91.50%
 Step: 730, Train Loss: 0.3048, Train Accuracy: 85.75%, Loss: 0.2148, Accuracy: 92.00%
 Step: 740, Train Loss: 0.3048, Train Accuracy: 85.75%, Loss: 0.2146, Accuracy: 91.50%
 Step: 750, Train Loss: 0.3048, Train Accuracy: 85.62%, Loss: 0.2151, Accuracy:

92.00%

Step: 760, Train Loss: 0.3047, Train Accuracy: 85.75%, Loss: 0.2147, Accuracy: 92.00%

Step: 770, Train Loss: 0.3047, Train Accuracy: 85.75%, Loss: 0.2149, Accuracy: 92.00%

Step: 780, Train Loss: 0.3047, Train Accuracy: 85.75%, Loss: 0.2147, Accuracy: 92.00%

Step: 790, Train Loss: 0.3047, Train Accuracy: 85.62%, Loss: 0.2144, Accuracy: 91.50%

Step: 800, Train Loss: 0.3047, Train Accuracy: 85.75%, Loss: 0.2143, Accuracy: 91.50%

Step: 810, Train Loss: 0.3046, Train Accuracy: 85.75%, Loss: 0.2145, Accuracy: 92.00%

Step: 820, Train Loss: 0.3046, Train Accuracy: 85.75%, Loss: 0.2145, Accuracy: 92.00%

Step: 830, Train Loss: 0.3046, Train Accuracy: 85.62%, Loss: 0.2146, Accuracy: 92.00%

Step: 840, Train Loss: 0.3046, Train Accuracy: 85.62%, Loss: 0.2146, Accuracy: 92.00%

Step: 850, Train Loss: 0.3046, Train Accuracy: 85.62%, Loss: 0.2144, Accuracy: 92.00%

Step: 860, Train Loss: 0.3046, Train Accuracy: 85.75%, Loss: 0.2145, Accuracy: 92.00%

Step: 870, Train Loss: 0.3045, Train Accuracy: 85.75%, Loss: 0.2142, Accuracy: 92.00%

Step: 880, Train Loss: 0.3046, Train Accuracy: 85.62%, Loss: 0.2147, Accuracy: 92.00%

Step: 890, Train Loss: 0.3045, Train Accuracy: 85.75%, Loss: 0.2143, Accuracy: 92.00%

Step: 900, Train Loss: 0.3046, Train Accuracy: 85.62%, Loss: 0.2147, Accuracy: 92.00%

Step: 910, Train Loss: 0.3045, Train Accuracy: 85.62%, Loss: 0.2145, Accuracy: 92.00%

Step: 920, Train Loss: 0.3045, Train Accuracy: 85.75%, Loss: 0.2144, Accuracy: 92.00%

Step: 930, Train Loss: 0.3045, Train Accuracy: 85.62%, Loss: 0.2142, Accuracy: 92.00%

Step: 940, Train Loss: 0.3045, Train Accuracy: 85.62%, Loss: 0.2142, Accuracy: 92.00%

Step: 950, Train Loss: 0.3045, Train Accuracy: 85.62%, Loss: 0.2142, Accuracy: 92.00%

Step: 960, Train Loss: 0.3044, Train Accuracy: 85.62%, Loss: 0.2141, Accuracy: 92.00%

Step: 970, Train Loss: 0.3044, Train Accuracy: 85.75%, Loss: 0.2142, Accuracy: 92.00%

Step: 980, Train Loss: 0.3044, Train Accuracy: 85.75%, Loss: 0.2141, Accuracy: 92.00%

Step: 990, Train Loss: 0.3044, Train Accuracy: 85.88%, Loss: 0.2138, Accuracy:

91.50%

Step: 1000, Train Loss: 0.3044, Train Accuracy: 85.62%, Loss: 0.2141, Accuracy: 92.00%

Step: 1010, Train Loss: 0.3044, Train Accuracy: 85.62%, Loss: 0.2143, Accuracy: 92.00%

Step: 1020, Train Loss: 0.3044, Train Accuracy: 85.62%, Loss: 0.2143, Accuracy: 92.00%

Step: 1030, Train Loss: 0.3044, Train Accuracy: 85.75%, Loss: 0.2138, Accuracy: 91.50%

Step: 1040, Train Loss: 0.3043, Train Accuracy: 85.75%, Loss: 0.2139, Accuracy: 92.00%

Step: 1050, Train Loss: 0.3044, Train Accuracy: 85.62%, Loss: 0.2142, Accuracy: 92.00%

Step: 1060, Train Loss: 0.3043, Train Accuracy: 85.75%, Loss: 0.2141, Accuracy: 92.00%

Step: 1070, Train Loss: 0.3043, Train Accuracy: 85.75%, Loss: 0.2139, Accuracy: 92.00%

Step: 1080, Train Loss: 0.3043, Train Accuracy: 85.75%, Loss: 0.2138, Accuracy: 91.50%

Step: 1090, Train Loss: 0.3043, Train Accuracy: 85.75%, Loss: 0.2139, Accuracy: 92.00%

Step: 1100, Train Loss: 0.3043, Train Accuracy: 85.75%, Loss: 0.2139, Accuracy: 92.00%

Step: 1110, Train Loss: 0.3043, Train Accuracy: 85.62%, Loss: 0.2140, Accuracy: 92.00%

Step: 1120, Train Loss: 0.3043, Train Accuracy: 85.62%, Loss: 0.2139, Accuracy: 92.00%

Step: 1130, Train Loss: 0.3043, Train Accuracy: 85.75%, Loss: 0.2139, Accuracy: 92.00%

Step: 1140, Train Loss: 0.3043, Train Accuracy: 85.62%, Loss: 0.2136, Accuracy: 91.50%

Step: 1150, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2139, Accuracy: 92.00%

Step: 1160, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2138, Accuracy: 92.00%

Step: 1170, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2140, Accuracy: 92.00%

Step: 1180, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2136, Accuracy: 91.50%

Step: 1190, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2137, Accuracy: 91.50%

Step: 1200, Train Loss: 0.3042, Train Accuracy: 85.62%, Loss: 0.2139, Accuracy: 92.00%

Step: 1210, Train Loss: 0.3042, Train Accuracy: 85.62%, Loss: 0.2138, Accuracy: 92.00%

Step: 1220, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2138, Accuracy: 92.00%

Step: 1230, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2137, Accuracy:

92.00%

Step: 1240, Train Loss: 0.3042, Train Accuracy: 85.62%, Loss: 0.2135, Accuracy: 91.50%

Step: 1250, Train Loss: 0.3041, Train Accuracy: 85.75%, Loss: 0.2137, Accuracy: 92.00%

Step: 1260, Train Loss: 0.3041, Train Accuracy: 85.75%, Loss: 0.2141, Accuracy: 92.00%

Step: 1270, Train Loss: 0.3042, Train Accuracy: 85.75%, Loss: 0.2135, Accuracy: 91.50%

Step: 1280, Train Loss: 0.3041, Train Accuracy: 85.75%, Loss: 0.2137, Accuracy: 92.00%

Step: 1290, Train Loss: 0.3041, Train Accuracy: 85.75%, Loss: 0.2137, Accuracy: 92.00%

Step: 1300, Train Loss: 0.3041, Train Accuracy: 85.62%, Loss: 0.2138, Accuracy: 92.00%

Step: 1310, Train Loss: 0.3041, Train Accuracy: 85.62%, Loss: 0.2137, Accuracy: 92.00%

Step: 1320, Train Loss: 0.3041, Train Accuracy: 85.75%, Loss: 0.2137, Accuracy: 92.00%

Step: 1330, Train Loss: 0.3041, Train Accuracy: 85.75%, Loss: 0.2139, Accuracy: 92.00%

Step: 1340, Train Loss: 0.3041, Train Accuracy: 85.75%, Loss: 0.2136, Accuracy: 91.50%

Step: 1350, Train Loss: 0.3040, Train Accuracy: 85.62%, Loss: 0.2138, Accuracy: 92.00%

Step: 1360, Train Loss: 0.3040, Train Accuracy: 85.75%, Loss: 0.2138, Accuracy: 92.00%

Step: 1370, Train Loss: 0.3041, Train Accuracy: 85.62%, Loss: 0.2141, Accuracy: 92.00%

Step: 1380, Train Loss: 0.3040, Train Accuracy: 85.62%, Loss: 0.2138, Accuracy: 92.00%

Step: 1390, Train Loss: 0.3040, Train Accuracy: 85.62%, Loss: 0.2138, Accuracy: 92.00%

Step: 1400, Train Loss: 0.3040, Train Accuracy: 85.62%, Loss: 0.2138, Accuracy: 92.00%

Step: 1410, Train Loss: 0.3040, Train Accuracy: 85.62%, Loss: 0.2137, Accuracy: 92.00%

Step: 1420, Train Loss: 0.3040, Train Accuracy: 85.75%, Loss: 0.2138, Accuracy: 92.00%

Step: 1430, Train Loss: 0.3040, Train Accuracy: 85.75%, Loss: 0.2138, Accuracy: 92.00%

Step: 1440, Train Loss: 0.3039, Train Accuracy: 85.75%, Loss: 0.2136, Accuracy: 92.00%

Step: 1450, Train Loss: 0.3039, Train Accuracy: 85.75%, Loss: 0.2135, Accuracy: 91.50%

Step: 1460, Train Loss: 0.3039, Train Accuracy: 85.62%, Loss: 0.2137, Accuracy: 92.00%

Step: 1470, Train Loss: 0.3039, Train Accuracy: 85.75%, Loss: 0.2140, Accuracy: 92.00%
 Step: 1480, Train Loss: 0.3039, Train Accuracy: 85.75%, Loss: 0.2135, Accuracy: 92.00%
 Step: 1490, Train Loss: 0.3039, Train Accuracy: 85.62%, Loss: 0.2138, Accuracy: 92.00%
 Step: 1499, Train Loss: 0.3039, Train Accuracy: 85.62%, Loss: 0.2137, Accuracy: 92.00%
 Training complete!
 Step: 0, Train Loss: 0.6929, Train Accuracy: 79.50%, Loss: 0.6929, Accuracy: 82.00%
 Step: 10, Train Loss: 0.6929, Train Accuracy: 80.12%, Loss: 0.6929, Accuracy: 83.00%
 Step: 20, Train Loss: 0.6929, Train Accuracy: 80.38%, Loss: 0.6929, Accuracy: 83.50%
 Step: 30, Train Loss: 0.6929, Train Accuracy: 80.62%, Loss: 0.6928, Accuracy: 83.00%
 Step: 40, Train Loss: 0.6928, Train Accuracy: 80.50%, Loss: 0.6928, Accuracy: 83.50%
 Step: 50, Train Loss: 0.6928, Train Accuracy: 80.75%, Loss: 0.6928, Accuracy: 84.00%
 Step: 60, Train Loss: 0.6928, Train Accuracy: 81.00%, Loss: 0.6927, Accuracy: 84.00%
 Step: 70, Train Loss: 0.6927, Train Accuracy: 81.00%, Loss: 0.6927, Accuracy: 84.00%
 Step: 80, Train Loss: 0.6927, Train Accuracy: 80.88%, Loss: 0.6926, Accuracy: 84.00%
 Step: 90, Train Loss: 0.6926, Train Accuracy: 80.88%, Loss: 0.6926, Accuracy: 84.00%
 Step: 100, Train Loss: 0.6926, Train Accuracy: 81.00%, Loss: 0.6926, Accuracy: 84.00%
 Step: 110, Train Loss: 0.6926, Train Accuracy: 80.88%, Loss: 0.6925, Accuracy: 84.00%
 Step: 120, Train Loss: 0.6925, Train Accuracy: 80.75%, Loss: 0.6925, Accuracy: 83.50%
 Step: 130, Train Loss: 0.6925, Train Accuracy: 80.50%, Loss: 0.6924, Accuracy: 81.50%
 Step: 140, Train Loss: 0.6924, Train Accuracy: 80.62%, Loss: 0.6924, Accuracy: 82.00%
 Step: 150, Train Loss: 0.6924, Train Accuracy: 80.25%, Loss: 0.6923, Accuracy: 81.50%
 Step: 160, Train Loss: 0.6923, Train Accuracy: 79.75%, Loss: 0.6922, Accuracy: 81.00%
 Step: 170, Train Loss: 0.6922, Train Accuracy: 79.38%, Loss: 0.6922, Accuracy: 81.00%
 Step: 180, Train Loss: 0.6922, Train Accuracy: 79.25%, Loss: 0.6921, Accuracy: 80.00%
 Step: 190, Train Loss: 0.6921, Train Accuracy: 79.12%, Loss: 0.6920, Accuracy:

79.50%

Step: 200, Train Loss: 0.6920, Train Accuracy: 79.00%, Loss: 0.6919, Accuracy: 79.50%

Step: 210, Train Loss: 0.6920, Train Accuracy: 79.00%, Loss: 0.6919, Accuracy: 79.50%

Step: 220, Train Loss: 0.6919, Train Accuracy: 79.00%, Loss: 0.6918, Accuracy: 79.50%

Step: 230, Train Loss: 0.6918, Train Accuracy: 79.00%, Loss: 0.6917, Accuracy: 79.50%

Step: 240, Train Loss: 0.6917, Train Accuracy: 79.00%, Loss: 0.6916, Accuracy: 79.50%

Step: 250, Train Loss: 0.6916, Train Accuracy: 79.00%, Loss: 0.6915, Accuracy: 79.00%

Step: 260, Train Loss: 0.6915, Train Accuracy: 79.00%, Loss: 0.6914, Accuracy: 79.00%

Step: 270, Train Loss: 0.6914, Train Accuracy: 79.00%, Loss: 0.6912, Accuracy: 79.00%

Step: 280, Train Loss: 0.6912, Train Accuracy: 79.00%, Loss: 0.6911, Accuracy: 79.00%

Step: 290, Train Loss: 0.6911, Train Accuracy: 79.00%, Loss: 0.6910, Accuracy: 79.00%

Step: 300, Train Loss: 0.6910, Train Accuracy: 79.00%, Loss: 0.6908, Accuracy: 79.00%

Step: 310, Train Loss: 0.6908, Train Accuracy: 79.00%, Loss: 0.6907, Accuracy: 79.00%

Step: 320, Train Loss: 0.6907, Train Accuracy: 79.12%, Loss: 0.6905, Accuracy: 79.00%

Step: 330, Train Loss: 0.6905, Train Accuracy: 79.12%, Loss: 0.6903, Accuracy: 78.50%

Step: 340, Train Loss: 0.6903, Train Accuracy: 79.12%, Loss: 0.6901, Accuracy: 78.00%

Step: 350, Train Loss: 0.6901, Train Accuracy: 79.12%, Loss: 0.6899, Accuracy: 78.00%

Step: 360, Train Loss: 0.6899, Train Accuracy: 79.12%, Loss: 0.6897, Accuracy: 78.00%

Step: 370, Train Loss: 0.6897, Train Accuracy: 79.12%, Loss: 0.6895, Accuracy: 78.50%

Step: 380, Train Loss: 0.6895, Train Accuracy: 79.12%, Loss: 0.6892, Accuracy: 78.50%

Step: 390, Train Loss: 0.6893, Train Accuracy: 79.12%, Loss: 0.6890, Accuracy: 78.50%

Step: 400, Train Loss: 0.6890, Train Accuracy: 79.12%, Loss: 0.6887, Accuracy: 78.50%

Step: 410, Train Loss: 0.6887, Train Accuracy: 79.12%, Loss: 0.6884, Accuracy: 78.50%

Step: 420, Train Loss: 0.6884, Train Accuracy: 79.12%, Loss: 0.6881, Accuracy: 78.00%

Step: 430, Train Loss: 0.6881, Train Accuracy: 79.12%, Loss: 0.6877, Accuracy:

78.00%
Step: 440, Train Loss: 0.6878, Train Accuracy: 79.12%, Loss: 0.6874, Accuracy: 78.00%
Step: 450, Train Loss: 0.6874, Train Accuracy: 79.12%, Loss: 0.6870, Accuracy: 78.00%
Step: 460, Train Loss: 0.6870, Train Accuracy: 79.12%, Loss: 0.6866, Accuracy: 78.00%
Step: 470, Train Loss: 0.6866, Train Accuracy: 79.12%, Loss: 0.6862, Accuracy: 78.00%
Step: 480, Train Loss: 0.6862, Train Accuracy: 79.12%, Loss: 0.6857, Accuracy: 78.00%
Step: 490, Train Loss: 0.6858, Train Accuracy: 79.12%, Loss: 0.6852, Accuracy: 78.00%
Step: 500, Train Loss: 0.6853, Train Accuracy: 79.12%, Loss: 0.6847, Accuracy: 78.00%
Step: 510, Train Loss: 0.6848, Train Accuracy: 79.12%, Loss: 0.6842, Accuracy: 78.00%
Step: 520, Train Loss: 0.6842, Train Accuracy: 79.12%, Loss: 0.6836, Accuracy: 78.00%
Step: 530, Train Loss: 0.6836, Train Accuracy: 79.12%, Loss: 0.6829, Accuracy: 78.00%
Step: 540, Train Loss: 0.6830, Train Accuracy: 79.12%, Loss: 0.6823, Accuracy: 78.00%
Step: 550, Train Loss: 0.6824, Train Accuracy: 79.12%, Loss: 0.6816, Accuracy: 78.00%
Step: 560, Train Loss: 0.6817, Train Accuracy: 79.12%, Loss: 0.6808, Accuracy: 78.00%
Step: 570, Train Loss: 0.6810, Train Accuracy: 79.12%, Loss: 0.6801, Accuracy: 78.00%
Step: 580, Train Loss: 0.6802, Train Accuracy: 79.12%, Loss: 0.6792, Accuracy: 78.00%
Step: 590, Train Loss: 0.6794, Train Accuracy: 79.12%, Loss: 0.6783, Accuracy: 78.00%
Step: 600, Train Loss: 0.6785, Train Accuracy: 79.12%, Loss: 0.6774, Accuracy: 78.00%
Step: 610, Train Loss: 0.6776, Train Accuracy: 79.12%, Loss: 0.6764, Accuracy: 78.00%
Step: 620, Train Loss: 0.6766, Train Accuracy: 79.12%, Loss: 0.6754, Accuracy: 78.00%
Step: 630, Train Loss: 0.6756, Train Accuracy: 79.12%, Loss: 0.6742, Accuracy: 78.00%
Step: 640, Train Loss: 0.6745, Train Accuracy: 79.12%, Loss: 0.6731, Accuracy: 78.00%
Step: 650, Train Loss: 0.6733, Train Accuracy: 79.12%, Loss: 0.6718, Accuracy: 78.00%
Step: 660, Train Loss: 0.6721, Train Accuracy: 79.12%, Loss: 0.6705, Accuracy: 78.00%
Step: 670, Train Loss: 0.6708, Train Accuracy: 79.12%, Loss: 0.6691, Accuracy:

78.00%

Step: 680, Train Loss: 0.6695, Train Accuracy: 79.12%, Loss: 0.6677, Accuracy: 78.00%

Step: 690, Train Loss: 0.6680, Train Accuracy: 79.12%, Loss: 0.6662, Accuracy: 78.00%

Step: 700, Train Loss: 0.6665, Train Accuracy: 79.12%, Loss: 0.6646, Accuracy: 78.00%

Step: 710, Train Loss: 0.6650, Train Accuracy: 79.12%, Loss: 0.6629, Accuracy: 78.00%

Step: 720, Train Loss: 0.6633, Train Accuracy: 79.12%, Loss: 0.6611, Accuracy: 78.00%

Step: 730, Train Loss: 0.6616, Train Accuracy: 79.12%, Loss: 0.6592, Accuracy: 78.00%

Step: 740, Train Loss: 0.6598, Train Accuracy: 79.12%, Loss: 0.6572, Accuracy: 78.50%

Step: 750, Train Loss: 0.6579, Train Accuracy: 79.12%, Loss: 0.6552, Accuracy: 78.50%

Step: 760, Train Loss: 0.6559, Train Accuracy: 79.12%, Loss: 0.6530, Accuracy: 78.50%

Step: 770, Train Loss: 0.6538, Train Accuracy: 79.12%, Loss: 0.6508, Accuracy: 78.50%

Step: 780, Train Loss: 0.6516, Train Accuracy: 79.12%, Loss: 0.6484, Accuracy: 78.50%

Step: 790, Train Loss: 0.6493, Train Accuracy: 79.12%, Loss: 0.6459, Accuracy: 78.50%

Step: 800, Train Loss: 0.6470, Train Accuracy: 79.12%, Loss: 0.6434, Accuracy: 78.50%

Step: 810, Train Loss: 0.6445, Train Accuracy: 79.12%, Loss: 0.6407, Accuracy: 78.00%

Step: 820, Train Loss: 0.6419, Train Accuracy: 79.12%, Loss: 0.6379, Accuracy: 78.00%

Step: 830, Train Loss: 0.6393, Train Accuracy: 79.12%, Loss: 0.6350, Accuracy: 78.50%

Step: 840, Train Loss: 0.6365, Train Accuracy: 79.12%, Loss: 0.6320, Accuracy: 79.00%

Step: 850, Train Loss: 0.6337, Train Accuracy: 79.25%, Loss: 0.6289, Accuracy: 79.00%

Step: 860, Train Loss: 0.6307, Train Accuracy: 79.12%, Loss: 0.6257, Accuracy: 79.00%

Step: 870, Train Loss: 0.6277, Train Accuracy: 79.00%, Loss: 0.6224, Accuracy: 79.00%

Step: 880, Train Loss: 0.6245, Train Accuracy: 79.00%, Loss: 0.6190, Accuracy: 79.00%

Step: 890, Train Loss: 0.6213, Train Accuracy: 79.00%, Loss: 0.6154, Accuracy: 79.00%

Step: 900, Train Loss: 0.6180, Train Accuracy: 79.00%, Loss: 0.6118, Accuracy: 79.00%

Step: 910, Train Loss: 0.6145, Train Accuracy: 79.00%, Loss: 0.6080, Accuracy:

79.50%

Step: 920, Train Loss: 0.6110, Train Accuracy: 79.00%, Loss: 0.6042, Accuracy: 79.50%

Step: 930, Train Loss: 0.6074, Train Accuracy: 79.00%, Loss: 0.6002, Accuracy: 79.50%

Step: 940, Train Loss: 0.6038, Train Accuracy: 79.00%, Loss: 0.5962, Accuracy: 79.50%

Step: 950, Train Loss: 0.6000, Train Accuracy: 79.00%, Loss: 0.5921, Accuracy: 79.50%

Step: 960, Train Loss: 0.5962, Train Accuracy: 79.00%, Loss: 0.5879, Accuracy: 79.50%

Step: 970, Train Loss: 0.5923, Train Accuracy: 79.00%, Loss: 0.5836, Accuracy: 79.50%

Step: 980, Train Loss: 0.5884, Train Accuracy: 79.12%, Loss: 0.5792, Accuracy: 79.50%

Step: 990, Train Loss: 0.5844, Train Accuracy: 79.25%, Loss: 0.5748, Accuracy: 79.50%

Step: 1000, Train Loss: 0.5803, Train Accuracy: 79.38%, Loss: 0.5703, Accuracy: 79.50%

Step: 1010, Train Loss: 0.5762, Train Accuracy: 79.25%, Loss: 0.5657, Accuracy: 79.50%

Step: 1020, Train Loss: 0.5721, Train Accuracy: 79.25%, Loss: 0.5611, Accuracy: 79.50%

Step: 1030, Train Loss: 0.5679, Train Accuracy: 79.38%, Loss: 0.5564, Accuracy: 80.00%

Step: 1040, Train Loss: 0.5637, Train Accuracy: 79.62%, Loss: 0.5517, Accuracy: 80.50%

Step: 1050, Train Loss: 0.5594, Train Accuracy: 79.62%, Loss: 0.5470, Accuracy: 81.00%

Step: 1060, Train Loss: 0.5552, Train Accuracy: 79.62%, Loss: 0.5422, Accuracy: 81.00%

Step: 1070, Train Loss: 0.5509, Train Accuracy: 79.88%, Loss: 0.5374, Accuracy: 81.00%

Step: 1080, Train Loss: 0.5466, Train Accuracy: 79.88%, Loss: 0.5325, Accuracy: 81.50%

Step: 1090, Train Loss: 0.5423, Train Accuracy: 80.12%, Loss: 0.5277, Accuracy: 81.50%

Step: 1100, Train Loss: 0.5379, Train Accuracy: 80.00%, Loss: 0.5228, Accuracy: 82.00%

Step: 1110, Train Loss: 0.5336, Train Accuracy: 80.25%, Loss: 0.5179, Accuracy: 82.50%

Step: 1120, Train Loss: 0.5293, Train Accuracy: 80.12%, Loss: 0.5130, Accuracy: 82.50%

Step: 1130, Train Loss: 0.5250, Train Accuracy: 80.25%, Loss: 0.5081, Accuracy: 83.00%

Step: 1140, Train Loss: 0.5207, Train Accuracy: 80.50%, Loss: 0.5032, Accuracy: 83.00%

Step: 1150, Train Loss: 0.5165, Train Accuracy: 80.50%, Loss: 0.4983, Accuracy:

83.50%

Step: 1160, Train Loss: 0.5122, Train Accuracy: 80.38%, Loss: 0.4935, Accuracy: 83.50%

Step: 1170, Train Loss: 0.5080, Train Accuracy: 80.50%, Loss: 0.4886, Accuracy: 83.50%

Step: 1180, Train Loss: 0.5038, Train Accuracy: 80.38%, Loss: 0.4838, Accuracy: 83.50%

Step: 1190, Train Loss: 0.4996, Train Accuracy: 80.62%, Loss: 0.4790, Accuracy: 84.50%

Step: 1200, Train Loss: 0.4955, Train Accuracy: 80.75%, Loss: 0.4742, Accuracy: 84.50%

Step: 1210, Train Loss: 0.4914, Train Accuracy: 80.75%, Loss: 0.4695, Accuracy: 85.00%

Step: 1220, Train Loss: 0.4874, Train Accuracy: 81.00%, Loss: 0.4648, Accuracy: 85.50%

Step: 1230, Train Loss: 0.4834, Train Accuracy: 81.00%, Loss: 0.4601, Accuracy: 85.50%

Step: 1240, Train Loss: 0.4794, Train Accuracy: 81.12%, Loss: 0.4555, Accuracy: 85.50%

Step: 1250, Train Loss: 0.4755, Train Accuracy: 81.12%, Loss: 0.4509, Accuracy: 86.50%

Step: 1260, Train Loss: 0.4717, Train Accuracy: 81.25%, Loss: 0.4464, Accuracy: 87.00%

Step: 1270, Train Loss: 0.4679, Train Accuracy: 81.38%, Loss: 0.4419, Accuracy: 87.00%

Step: 1280, Train Loss: 0.4642, Train Accuracy: 81.50%, Loss: 0.4375, Accuracy: 87.00%

Step: 1290, Train Loss: 0.4605, Train Accuracy: 81.75%, Loss: 0.4331, Accuracy: 87.00%

Step: 1300, Train Loss: 0.4569, Train Accuracy: 81.75%, Loss: 0.4288, Accuracy: 87.00%

Step: 1310, Train Loss: 0.4533, Train Accuracy: 81.88%, Loss: 0.4245, Accuracy: 87.00%

Step: 1320, Train Loss: 0.4498, Train Accuracy: 81.88%, Loss: 0.4203, Accuracy: 87.00%

Step: 1330, Train Loss: 0.4464, Train Accuracy: 82.00%, Loss: 0.4162, Accuracy: 87.00%

Step: 1340, Train Loss: 0.4430, Train Accuracy: 82.00%, Loss: 0.4122, Accuracy: 87.00%

Step: 1350, Train Loss: 0.4397, Train Accuracy: 82.00%, Loss: 0.4082, Accuracy: 87.00%

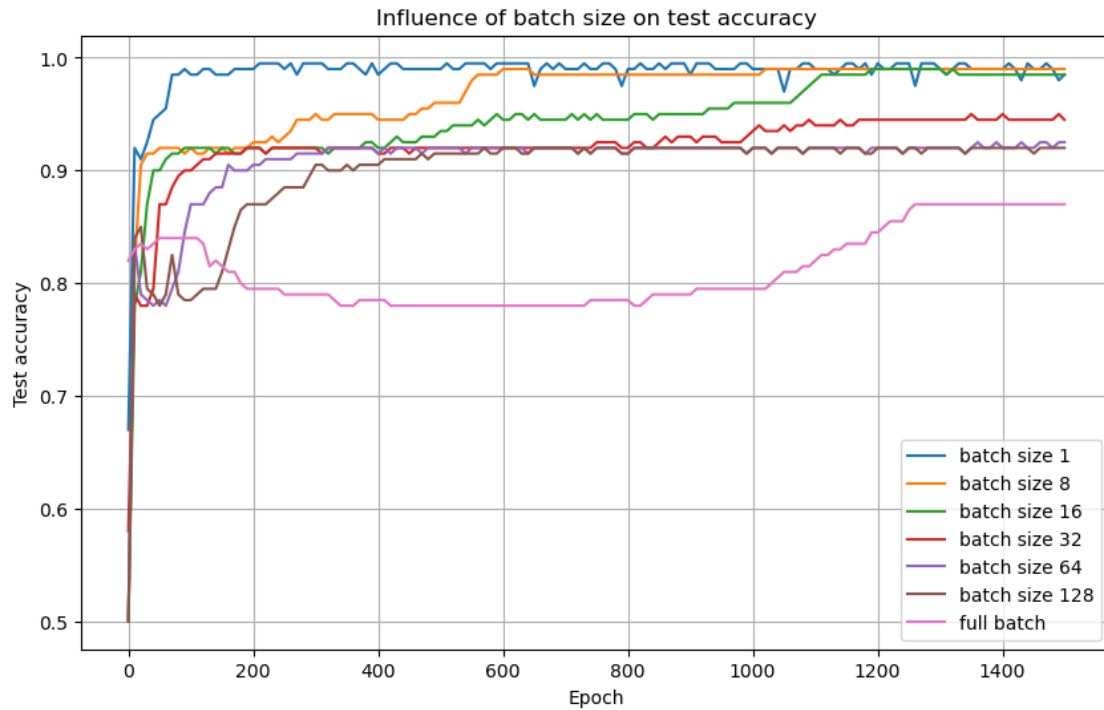
Step: 1360, Train Loss: 0.4364, Train Accuracy: 82.00%, Loss: 0.4042, Accuracy: 87.00%

Step: 1370, Train Loss: 0.4332, Train Accuracy: 82.00%, Loss: 0.4004, Accuracy: 87.00%

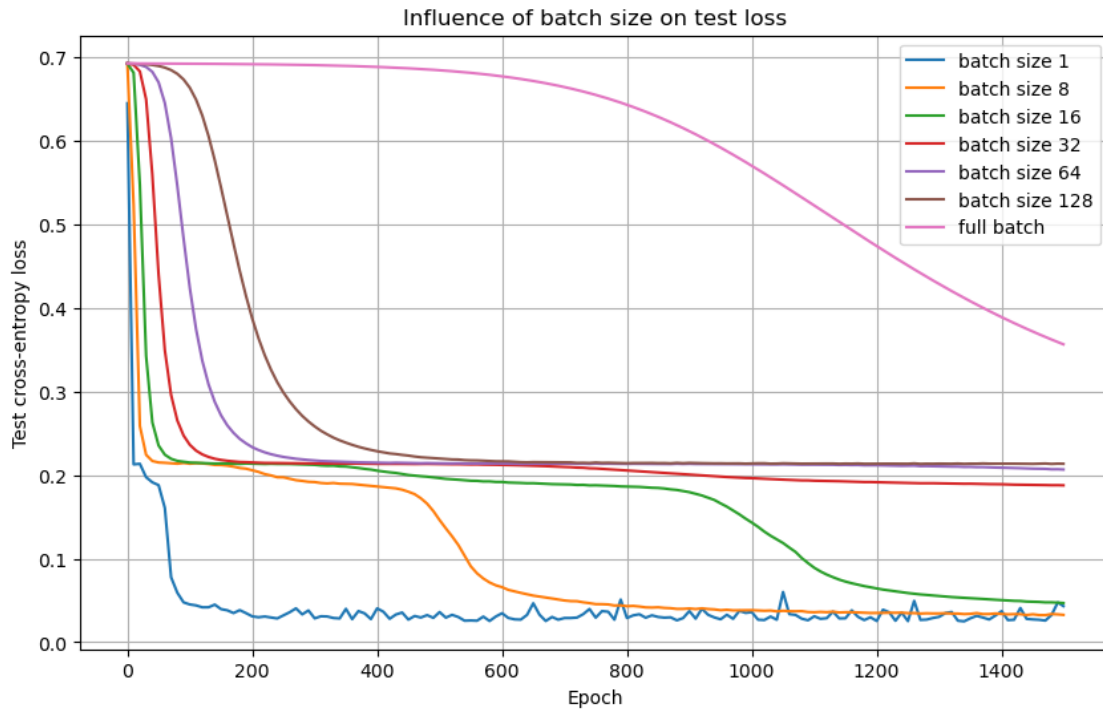
Step: 1380, Train Loss: 0.4301, Train Accuracy: 82.00%, Loss: 0.3966, Accuracy: 87.00%

Step: 1390, Train Loss: 0.4271, Train Accuracy: 82.12%, Loss: 0.3928, Accuracy: 87.00%
Step: 1400, Train Loss: 0.4241, Train Accuracy: 82.12%, Loss: 0.3892, Accuracy: 87.00%
Step: 1410, Train Loss: 0.4211, Train Accuracy: 82.25%, Loss: 0.3856, Accuracy: 87.00%
Step: 1420, Train Loss: 0.4183, Train Accuracy: 82.50%, Loss: 0.3820, Accuracy: 87.00%
Step: 1430, Train Loss: 0.4155, Train Accuracy: 82.50%, Loss: 0.3786, Accuracy: 87.00%
Step: 1440, Train Loss: 0.4128, Train Accuracy: 82.50%, Loss: 0.3752, Accuracy: 87.00%
Step: 1450, Train Loss: 0.4101, Train Accuracy: 82.50%, Loss: 0.3719, Accuracy: 87.00%
Step: 1460, Train Loss: 0.4075, Train Accuracy: 82.50%, Loss: 0.3686, Accuracy: 87.00%
Step: 1470, Train Loss: 0.4050, Train Accuracy: 82.50%, Loss: 0.3655, Accuracy: 87.00%
Step: 1480, Train Loss: 0.4025, Train Accuracy: 82.62%, Loss: 0.3624, Accuracy: 87.00%
Step: 1490, Train Loss: 0.4001, Train Accuracy: 82.50%, Loss: 0.3593, Accuracy: 87.00%
Step: 1499, Train Loss: 0.3980, Train Accuracy: 82.50%, Loss: 0.3566, Accuracy: 87.00%
Training complete!

```
[6]: plt.figure(figsize=(10, 6))
      for label, history in batch_results.items():
          plt.plot(history["steps"], history["test_accuracy"], label=label)
      plt.xlabel("Epoch")
      plt.ylabel("Test accuracy")
      plt.title("Influence of batch size on test accuracy")
      plt.legend()
      plt.grid(True)
      plt.show()
```



```
[7]: plt.figure(figsize=(10, 6))
for label, history in batch_results.items():
    plt.plot(history["steps"], history["test_loss"], label=label)
plt.xlabel("Epoch")
plt.ylabel("Test cross-entropy loss")
plt.title("Influence of batch size on test loss")
plt.legend()
plt.grid(True)
plt.show()
```



```
[8]: summary = {
    label: history["test_accuracy"][-1]
    for label, history in batch_results.items()
}
for label, final_accuracy in summary.items():
    print(f"{label}: {final_accuracy:.2%}")
```

```
batch size 1: 98.50%
batch size 8: 99.00%
batch size 16: 98.50%
batch size 32: 94.50%
batch size 64: 92.50%
batch size 128: 92.00%
full batch: 87.00%
```